

Dynamic yet Persistent: Investigating Digital Traces of Human Behaviour

Sara Heydari



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Abstract

We are citizens of the digital age, an age in which our smallest everyday actions leave digital traces behind. These traces can be used to reconstruct and analyse the patterns of human behaviour at both individual and societal levels. The growing abundance of quantitative data on almost every aspect of human behaviour is driving a paradigm shift from traditional social sciences to computational social sciences.

This thesis utilises a wide range of digital traces of human behaviour to study and model the structure of social networks, as well as population-level patterns of commuting and travelling. A longitudinal study of these patterns reveals that while both personal social networks and population commuting networks are dynamic and undergo gradual changes and abrupt external disturbances, they also exhibit persistence in certain aspects and retain some of their distinctive features.

Our work advances the knowledge of personal networks by identifying their universal and individual features through the study of a large population and multiple communication media. To clarify the generative mechanisms behind the observed universal patterns and individual variations within them, we present an ego-network model that connects the structure of ego-networks to the communication strategies of egos.

Furthermore, we utilise rich and high-resolution digital communication logs and computational methods to validate the predictions of sociological theories that predate the emergence of computational social science as a field. Particularly, we show that the temporal manifestation of social ties can indicate the extent of their multiplexity (i.e., whether multiple social contexts underlie the tie) and that the multiplexity of ties impacts their functional features, such as their role in network connectivity.

In addition to social networks, this thesis enriches our understanding of human mobility patterns by building predictive models that employ digital traces data. These models are capable of real-time estimation of country-wide mobility flows, even amidst rapid and significant distortions in mobility patterns.

Taken together, my results highlight the potential of digital trace data and computational methods for advancing our understanding of various aspects of human behaviour. It applies these insights to predict human behaviour, identify universal and individual human traits, and contribute to bridging the new field of computational social science with the rich tradition of classical social science.

Keywords Social Networks, Ego-networks, Social Signatures, Human Mobility

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Preface

This Thesis marks the end of a long journey that probably wouldn't have been possible – and surely not the same – without the support of the countless people involved.

First, I would like to thank my supervisor, Jari Saramäki, for his invaluable support and guidance through this journey. Thanks for all the opportunities you provided me to learn and grow, for giving me the freedom to wander around in my research, and for always offering your help when things were getting difficult. My gratitude also goes to János Kertész for hosting my research visit at Central European University, which ignited an amazing collaboration with János and Gerardo Iñiguez. Thanks, János, for generously offering your insight, knowledge, and support. And thanks, Gerardo, for all the enthusiasm you put into our scientific discussions, whether walking me through your mathematical art and craft or brainstorming about new research projects. I would also like to thank Mikko Kivelä, Talayeh Aledavood, Lasse Leskelä, Juhi Kulshrestha, and Petter Holme for supporting me through my academic journey.

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I cannot thank my friends and colleagues in the complex systems research group enough: Silja, Javier, Ana, Tarmo, Hasti, Arash, Onerva, Tuomas, Zahra, Sara, Elisa, Kiki, Äli, Ali, Abbas, Vesa, Arsi, and many others. Thanks for making my time at Aalto so special.

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I am also grateful to my friends at the Engareh reading club. Our

regular discussions on philosophy, history, and other topics were a great opportunity for me to practice critical thinking.

I would also like to thank all those who made me feel at home during these PhD years. Home is a complicated concept, and as an immigrant, I know just too well that it is not at a street address that you can find the home, but in all those small things that make you feel like you belong. Home can be in the carrying voices of the women on the streets in Tehran that reach me from thousands of miles away; or it can be in the peaceful silence that echoes over a Finnish lake glowing in the sunset; or in the scent of yellow marigold flowers that takes me all the way from a sunny day in Espoo to a breezy childhood evening at my grandparents' yard; or in gazing at the stars from the terrace of a village-house, while being wrapped in the warmth of a beautiful friendship. I want to thank all of those who created such moments with me: all those who hiked with me, who danced with me, who cooked with me, who laughed with me, who cried with me, who dreamed with me, and who loved with me.

Finally, I would like to thank my parents for giving me the freedom and courage to find my own way. Your love and support have always been the unwavering flame that warms my heart.

Espoo, July 31, 2024,

Sara Heydari

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List of Publications

This thesis consists of an overview and of the following publications which are referred to in the text by their Roman numerals.

- I** S. Heydari, S. G. Roberts, R. I. M. Dunbar and J. Saramäki. Multichannel social signatures and persistent features of ego networks. *Applied Network Science*, 3(1), Dec. 2018.
- II** G. Iñiguez, S. Heydari, J. Kertész and J. Saramäki. Universal patterns in egocentric communication networks. *Nature Communications*, 14(1), Aug. 2023.
- III** S. Heydari, G. Iñiguez, J. Kertész and J. Saramäki. Disentangling degree and tie strength heterogeneity in egocentric social networks. Submitted to *Peer Review*, Apr. 2024.
- IV** J. Ureña-Carrion, S. Heydari, T. Aledavood, J. Saramäki, and M. Kivelä. Multiplexity is temporal: effects of social times on network structure. Submitted to *Peer Review*, March 2024.
- V** S. Heydari, Z. Huang, T. Hiraoka, A. P. L. Chávez, T. Ala-Nissila, L. Leskelä, M. Kivelä and J. Saramäki. Estimating inter-regional mobility during disruption: comparing and combining different data sources. *Travel Behaviour and Society*, 31, 93–105, Apr. 2023.

Author's contributions

Publication I: “Multichannel social signatures and persistent features of ego networks”

S.H. is the main author of the article. S.G.R. and R.I.M.D. conceived and performed the data collection experiment. S.H. and J.S. analyzed data. S.H. developed the code for processing the data. S.H., J.S., S.G.R., and R.I.M.D. wrote the paper. All authors read and approved the final manuscript.

Publication II: “Universal patterns in egocentric communication networks”

G.I. is the main author of the article. G.I., S.H., J.K., and J.S. conceived, designed, and developed the study. G.I. and S.H. analyzed empirical data. G.I. derived analytical results and performed numerical simulations and model fitting. G.I., S.H., J.K., and J.S. wrote the paper.

Publication III: “Disentangling degree and tie strength heterogeneity in egocentric social networks”

S.H. is the main author of the article. S.H., G.I., J.K., and J.S. conceived, designed, and developed the study. S.H. analysed the empirical data. G.I. derived the analytical relations in Section S2.2. S.H., G.I., J.K., and J.S. wrote the paper.

Publication IV: “Multiplexity is temporal: effects of social times on network structure”

J.U.C is the main author of the article. All authors conceived the study. J.U.C. designed and developed the methods and analyses. J.U.C. implemented the models and empirical data studies, to which S.H. also contributed. All authors contributed to the interpretation of the results. J.U.C. wrote the first draft and all authors revised and approved the manuscript.

Publication V: “Estimating inter-regional mobility during disruption: comparing and combining different data sources”

S.H. is the main author of the article. All the authors except A.P.L.C. devised the main idea together. All the authors except A.P.L.C. contributed to conceiving the models. S.H. developed the codes for implementing the models and analysed their performances. All the authors except A.P.L.C. contributed to interpreting the results. L.L. wrote the mathematical proof for the asymmetric radiation model in Appendix C. A.P.L.C. preprocessed the road traffic data. S.H. produced all the plots except Figures 1 and S1, which Z.H. illustrated. S.H. wrote the majority of the first draft, while Z.H. contributed to writing the introduction. all authors revised and approved the manuscript.

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Abbreviations

LCC Largest Connected Component

JSD Jensen-Shannon Divergence

ML Maximum Likelihood

NNMF Non-negative Matrix Factorization

GPS Global Positioning System

OD matrix Origin–Destination matrix

CDRs Call Detail Records

GOF Goodness-of-fit

SMS Short Message Service

MIT Massachusetts Institute of Technology

COVID-19 Coronavirus Disease 2019

CNS Copenhagen Network Studies

RFID tags Radio Frequency Identification tags

CCDF Complementary Cumulative Distribution Function

Symbols

k Node degree

O_{ij} Overlap for tie ij

e Ego

w_{ei} The i th most contacted alter of ego e

s_e Social signature of ego e

H Shannon entropy

d_{JSD} Jensen-Shannon distance

L^2 Euclidean distance

G Gini Coefficient

β Alter-preferentiality parameter

1. Social Science, from Comte to the digital age

At the beginning of the 19th century, when science had already enabled humankind to predict and harness the forces of nature, some scholars began to wonder if the same scientific methods could be extended to the study of society and human affairs [29]. Among them was August Comte, the founder of positivism, who in his book titled *Course of Positive Philosophy* stated:

Now that human mind has grasped celestial and terrestrial physics; mechanical and chemical; organic physics, both vegetables and animals; there remains one science, to fill up the series of sciences of observation: social physics. This is what men have now most need of [38].

Since then, the unification of sciences has been a topic of debate. On one end of the spectrum were scientists like Bertrand Russell, dreaming of “a mathematics of human behaviour as precise as the mathematics of machines” [135], while on the other end, interpretivists and antipositivists argued that the study of the social realm is intrinsically different from the study of natural phenomena [100], and thus the scientific methods cannot be extended to the study of human affairs but rather an epistemologically different approach is required [77]. Despite the decades-long and ongoing debate between positivists and interpretivists [67], in practice, scientific methods such as the hypothetical-deductive model are widely used in modern social sciences.

Nevertheless, some extra challenges hinder the progress of the social sciences compared to the natural sciences. Among these challenges are:

- **Subjects become reactive when they feel observed [136].** Field research is one of the standard methods of collecting data on human behaviour. This method involves careful observation of the behaviour of subjects in their natural settings and may involve interacting with them [143]. However, people usually change their behaviour when they feel observed. Social scientists refer to this phenomenon as *reactivity*. Reac-

tivity can also bias the data collected via questionnaires and interviews. People are not always honest and tend to portray a socially desirable impression of themselves [143].

- **Low accuracy in recalling past events** Human recall of past events is not accurate [19, 143]. *Recall bias* is not only limited to the events in the distant past but also to the recent events when the details, like the exact timings, are concerned.
- **Data size** Traditional methods for collecting data on interpersonal relationships include field studies, interviews, and questionnaires. These methods are intrinsically labour-intensive [85] and thus not scalable to large populations.

In this thesis, we use automatically-collected *digital traces data* to overcome these obstacles in studying human behaviour, in particular to gain a better understanding of the the structure of social networks and patterns of human mobility (more on digital trace data in Sec. 1.1). The key contribution of this Thesis is the advancement of our understanding of egocentric networks structure, in particular in the context of multiple digital communication channels. We also apply some of the egocentric techniques to study human mobility patterns at the population level. A summary of the contributions of the publications included in this Thesis, as well as the outline of the Thesis, is provided at the end of this chapter in Section 1.3.

1.1 Data-driven computational social science

Science advances whenever we can take something that was once invisible and make it visible; and this is now taking place with regard to social networks and social processes.[85]

– Jon Kleinberg

We live in a digital age. Technologies such as the Internet, the World Wide Web, and smartphones have transformed every aspect of our lives [49], and as an inevitable part of this transformation, all sorts of our actions and movements are being automatically tracked [85]. As we live our daily lives – using mobile phones, paying with credit cards, checking out our grocery shopping, using public transportation, driving on the highways, typing a query in a search box, opening a web page, and using social media – we leave *digital traces* behind us [94].

The availability of a variety of streams of digital traces, recording the smallest actions of billions of people, is comparable with the invention of

the telescope and the microscope, as it makes what was once invisible or hard to grasp exposed to scientific inquiry and provides the opportunity to study human behaviour at a scale and depth which was not possible before [104]. The abundance of data and the increasing availability of computational resources for analysing these data has resulted in the emergence of the interdisciplinary field of *computational social science* [136, 83, 140, 94, 85, 49].

Indeed, besides the emergence of the novel resource of digital trace data, the digitalisation of older resources and the adoption of automated processing techniques are playing pivotal roles in the ongoing paradigm shift within the social sciences. These advancements foster the accessibility and analysis of long-existing resources, thus reshaping the landscape of social science research [136, 49]. Some examples are historical archives or literary corpora which are nowadays digitalised and can be analysed with an unprecedented speed using natural language processing methods. Multimedia data are another category which has been analysed by social scientists before the digital age but now are much easier to study. For example, in 1979, when Erving Goffman was analyzing how men and women are portrayed differently in photographic advertisements [60], he had to analyse the printed magazine advertisements manually, whereas a modern-day computational social scientist can utilise artificial intelligence methods of image processing to conduct a similar study on a much larger dataset and in an incomparably shorter time.

Digital trace data circumvent some of the challenges that traditional methods of data collection face. As discussed in the previous section, one challenge in studying human behaviour is that people tend to change their behaviour when they feel observed. An advantage of digital trace data is that the data collection procedure is rather non-intrusive, and people are either unaware that they are being tracked or not disturbed by it [136]. Moreover, studying the traces that people leave behind instead of asking them to recall what they have done, takes the recalling bias out of the equation [136].

Another advantage of data collected via people's interactions with digital devices and infrastructure is their scalability to large populations and the potential to capture human behaviour with a high resolution which has been out of the reach of traditional methods such as surveys and field observations [136]. The availability of high-resolution data on the behaviour of millions of people can, among others, provide insight into the population-level temporal patterns of communication and mobility [86, 140, 4, 71]. Moreover, the large sample size usually guarantees the statistical significance of the observation and the detection of even the most subtle of the trends [136].

Furthermore, streams of digital traces are continuously produced and can be used for real-time estimates or *nowcast* of human behaviour [136].

This is especially useful for gaining real-time insight into the population response to unexpected events such as natural disasters [141, 30], which can help policy-makers make informed decisions based on the current state [136]. In Publication V, we utilised a combination of different types of digital traces to provide real-time estimates of the travel behaviour of people in Finland in the midst of the COVID-19 pandemic, which induced drastic changes in the population travel behaviour.

Despite all the opportunities that digital trace data brings, it has its own drawbacks. For instance, while the data collected via traditional methods such as surveys are *designed*, meaning that they are gathered with pre-specific purposes and according to the research questions, digital trace datasets are in comparison *organic* or *found* [141, 136]. In designed data, the ratio of information to data is high; in contrast, researchers often need to approach organic data with data-mining techniques, that is, search and analyse large volumes of data to extract useful and related information [64].

Moreover, another shortcoming of organic data is that even though the data can be large in size, it might not provide a representative sample of the population we intend to study. While in designed data, the sampled population are carefully chosen to represent the target population, the demographic composition of organic data is often not clear. For instance, even though social media data can provide real-time insight into the opinions of its users, the socio-demographic composition of the users is often unknown to the researchers, and thus, the observed trends cannot be generalised to the general population [141, 136]. Moreover, such data should be cautiously used in longitudinal studies because the user composition of platforms tends to change over time [136].

The digital age also provides the possibility of running digital experiments to analyse social phenomena. Digital platforms are embedded in the daily lives of digital-age citizens, providing a natural setting for researchers to make interventions and measure the possible behavioural effects. While analysing organic data can only reveal correlations, conducting intervention-based randomised controlled experiments enables the researchers to infer cause-and-effect relationships [136]. The study conducted by Restivo et al. [129] is an example of leveraging already-existing digital platforms for performing randomised controlled social experiments. They studied the effect of peer endorsement on the productivity of Wikipedia contributors by taking a subset of the top one-percentile active Wikipedia editors and randomly dividing them into the endorsed and the control groups. The endorsed group members then were rewarded barnstars¹ by the researchers. The researchers then tracked the contributions of both endorsed and control groups in the following 90 days and

¹Barnstar is a virtual award that can be given by any Wikipedia editor to other editors to praise their work.

concluded that receiving a barnstar increases productivity by an average of 60% as compared with the control group [129]. While using already existing platforms is easier, custom-made platforms may be needed for answering specific research questions.

An example is the paper published in 2010 [33] on the effects of social network structure on the diffusion of behaviour in a population. To this end, a custom online health social media was built which enabled the researcher to fully control the social network structure of each user. The experiment showed that health-related behaviours are more likely to spread in clustered networks as compared to random networks.

Essentially, the possibility of performing large-scale controlled experiments is closing another gap between natural sciences and social sciences. Before the age of computational social science, conducting laboratory experiments in social sciences was considered unpractical [100] or unappealing [53] as the mainstream of social scientists believed that the controlled and artificial labs produce unrealistic data, which are irrelevant for understanding the “real world”. However, as compared to traditional labs, digital platforms are a natural environment for modern citizens. These platforms enable social scientists to conduct large-scale controlled experiments in realistic settings.

To conclude, the digital age has revolutionised not only our everyday lifestyles but also the social sciences. We are observing (and maybe contributing to) a paradigm shift [90] towards computational social sciences fuelled by the emergence of high-resolution data on human behaviour, the availability of gigantic computational powers, and an increasing trend of interdisciplinary approaches to complex social problems [35].

1.2 Mobile phone datasets

Mobile phone datasets are an important category of digital trace data and have been used in all five articles included in this thesis. Thus, I briefly review different subcategories of mobile phone data as well as some example studies fuelled by them.

Mobile phones are great candidates when it comes to collecting detailed and large-scale behavioural data. They are widely used around the globe, with the number of cellular subscriptions surpassing the world’s population [151] and an estimated global smartphone penetration rate of 68% in 2022 [150]. Moreover, mobile phones are personal devices, so the data associated with each device/subscription usually captures the behavioural patterns of a single individual. Additionally, we tend to keep our smartphones nearby and carry them around, and considering their built-in arsenal of sensors, they can collect data not only about our communication behaviour but also about our environment and surroundings, such as our

location and the presence of nearby digital devices.

This thesis uses three categories of mobile phone datasets: i) data collected by teleoperator companies, including both call detail records datasets (CDRs) collected by teleoperators for billing purposes, and origin-destination matrices (OD matrices) inferred from the locations of cell towers that users are at each timestamp proximate to. ii) Datasets collected for research purposes via mobile phone applications specifically designed for collecting behavioural data. iii) Mobility trends aggregated data published by companies such as Google and Facebook derived from the movement patterns of their large pool of users.

1.2.1 Data collected by teleoperators

The significant development of mobile phones has transformed them from devices for calling and texting to multipurpose handheld computers enhanced with a variety of sensors used for almost everything. Thus, the digital traces that we leave behind while using them can portray a detailed picture of our habits and lifestyles. However, even a basic old-style phone carries a SIM card and thus can expose rich data on communication and movement of the users. Teleoperator companies record the metadata (initiator and receiver phone numbers, event type, timestamp, duration/length) associated with the calls and messages that their subscribers send and receive.

Anonymised CDRs datasets were among the first large-scale and detailed datasets that computational social scientists have studied. In addition to the metadata associated with communication events, sometimes, CDRs datasets include basic demographic information such as the age and gender of the users which enables the comparison of different demographic cohorts (e.g. [44, 119]). CDRs datasets can potentially have large sample sizes and thus enable the study of human behaviour at a societal level. For example, Onnela *et al.* conducted two of the pioneer studies on the structure of large-scale social networks based on a CDRs dataset containing 20% of the population of a European country [117, 116]. Moreover, the high temporal resolution of these datasets (events recorded by the precision of seconds) revealed the temporal characteristics of human communication such as daily and weekly rhythms and burstiness [82, 4, 140, 28, 172, 81]. CDRs datasets have been used in Publication I, Publication III, Publication II and Publication IV included in this thesis.

Additionally, teleoperator companies can track their users' movements based on the locations of cell towers that the phones are proximate to. In Publication V we use the data provided by the teleoperator company Telia to train and test our model for estimating the flux of people travelling between Finnish municipalities.

1.2.2 Mobile phone data collection studies

CDRs datasets, due to their large sample sizes, are a valuable resource for investigating population behavioural trends. Nonetheless, they offer limited insights into human behaviour compared to the data obtainable through the diverse range of sensors integrated into smartphones. Moreover, with the availability of many alternative calling and messaging services, a decreasing portion of communication occurs through traditional call and messaging services provided by teleoperators. These factors have prompted researchers to develop methodologies for gathering behavioural data directly from smartphone users.

The Massachusetts Institute of Technology (MIT) “Reality Mining” experiment [48], conducted in 2004, is an early example of such data collection endeavours. In the Reality Mining project, researchers tracked behavioural data of a hundred volunteer MIT staff and students using identical Nokia 6600 phones equipped with a custom mobile application developed by researchers from the University of Helsinki for gathering behavioural data [126]. The collected data encompassed call and SMS logs, cell tower IDs, logs of nearby Bluetooth devices, application usage patterns, and phone statuses (e.g. active, idle, charging). Additionally, the dataset was supplemented with a survey asking basic demographic information of the users as well as questions about their social relationships and health-related lifestyle.

A more recent example is the “SensibleDTU” project, conducted at the Technical University of Denmark in 2012-2013. In this project, akin to the Reality Mining experiment, data was gathered using uniform smartphones and supplementary surveys. Due to technological advancements, the collected data is more comprehensive compared to the Reality Mining project and includes GPS²-based location data, logs of detected WiFi devices, and the subjects’ activities on the Facebook social media platform [153]. A subset of the data gathered in this project was made available for public use in 2019 [138, 137] under the title Copenhagen Network Study (CNS). In Publication II and Publication III, we use the one-month-long call and SMS logs included in this public dataset.

1.2.3 Mobility trends data

In response to the COVID-19 pandemic, Facebook and Google independently published data on changes in the population mobility trends for a limited duration [52, 51, 101]. The two companies extrapolated these trends from their application usage data. These datasets report the proportional changes in mobility compared to a pre-pandemic baseline. It should be noted that even though both these companies have a large pool

²GPS stands for Global Positioning System.

of users, it is hard to assess whether the users represent a demographically representative sample of the population, as no demographic details were provided. In publication Publication V we investigate the utility of these two datasets for estimating the travelling patterns in Finland.

1.3 Outline and contribution

This thesis utilises a variety of digital trace data and mathematical methods to expand our understanding of the structure and evolution of social networks as well as population-level mobility patterns. Chapter 2 is an introduction to the field of social network analysis and provides the background reading for understanding Publication I, Publication II, Publication III, and Publication IV where we analyse social networks. Chapters 3 – 5 are dedicated to presenting the results of the five publications included in this Thesis.

Publication I, Publication II, and Publication III focus on the structure and evolution of personal social networks and are summarised in Chapter 3. Particularly:

- In Publication I, we provided evidence on the existence of *social signatures*. In other words, we showed that people (egos) tend to shape their personal social networks (ego-networks) in their own distinctive ways, and these shapes are persistent in time and similar across the two different communication channels of call and text. Prior to our work, Refs. [139, 32] reported the persistence of social signatures using small call log datasets of respectively 24 and 93 individuals. Publication I, replicated the same finding using a much larger mobile phone dataset including around half a million active users. Moreover, we showed that akin to call signatures, social signatures reconstructed from text message ego-networks are also persistent.
- Moreover, Publication I presented a method for constructing comparable call and text networks from data and showed that while egos use these two communication channels for contacting very different sets of people, call and text social signatures resemble each other in shape.
- In Publication II, we investigated the possible driving mechanisms that result in the emergence of distinct and stable social signatures by devising a generative model that connects the structure of ego-networks to the communication strategy of the egos. By fitting our model to data we determine a *preferentiality parameter* for each ego-network that captures the tendency of the ego in re-inforcing his ties based on two competing reinforcement strategies: (i) cumulative advantage, where ties that are

already strong get even stronger, and (ii) random choice of communication partners. We fitted our model to ego-networks constructed from multiple sets of data on social interactions between millions of people via a variety of communication channels and showed that independently of the communication medium, the majority of the people have heterogeneous ego-networks and reinforce their social ties cumulatively rather than uniformly.

- In Publication III, we revisited the concept of social signature persistence by disentangling the two structural characteristics of social signatures: length and tie strength heterogeneity. This is a crucial task because while we are interested to know if people have persistent and individual strategies in distributing their limited communication resources among their contacts (i.e., persistent tie strength heterogeneities), as shown in Publication III, the distance functions used in the literature for comparing ego-networks are sensitive to their length differences. In Publication III, we studied four distance functions used for comparing the structure of ego-networks and showed that two of the four are fairly insensitive to differences in network size and, thus, more suitable for comparing ego-networks of different sizes.

In publication Publication IV, we used digital trace data to study the structure of population-level social networks guided by the framework of Feld's theory on focused organisation of social ties [54]. The results of the publication are summarised in Chapter 4. In particular:

- We showed that even though digital trace data on human communication, such as call logs, typically lack any information on the context of the social ties, it is possible to infer the multiplexity of ties from their temporal expression (which is captured in communication logs with a high-resolution of seconds). In Publication IV, we devised a method for quantifying the *temporal multiplexity* of ties based on the timestamps of their communication logs.
- The results of Publication IV indicate that temporal multiplexity is closely related to Feld's notion of contextual multiplexity [54] as the temporally multiplex ties exhibit the features predicted by Feld's theory: they are both locally and globally bridging and thus essential for the connectivity and cohesiveness of social networks.

In addition to social networks, this thesis enriches our understanding of human mobility patterns by building high-performing models fuelled by digital traces data. The models developed in Publication V use an egocentric network approach for understanding the mobility flow and

are capable of real-time estimation of country-wide mobility flows, even amidst rapid and significant distortions in mobility patterns. The results of Publication V are summarised in Chapter 5.

2. Social network analysis

Social life is relational; it's only because, say, blacks and whites occupy particular kinds of patterns in networks in relation to each other that "race" becomes an important variable. [37]

– Randall Collins

Humans are social beings. From the moment of birth, an individual is embedded in an intertwined dynamic web of relationships, which, to a large extent, determines their perception of self and identity, as well as their perception of the world [34]. One's position in this web affects the opportunities they are provided and how they are perceived by others and treated by society. Considering the importance of social relationships, network science is a natural modelling framework for studying human affairs both at individual and societal levels, as it provides us with methods to comprehend and analyse them in a relational way (see Figure 2.1 for a schematic visualisation of some basic network concepts that are used throughout this Thesis). After all, as Granovetter put, “no part of social life can be properly analysed without seeing how it is fundamentally embedded in networks of social relations” [62].

Social network analysis, or the study of societal phenomena through the lens of networks, can be considered to date back to the 1930s. The Romanian psychiatrist Jacob Moreno drew some of the first social network visualizations to study relationships in groups and to track their evolution [21]. Moreno's *sociograms* depict individuals as geometrical shapes (e.g. circles) and the relationships as lines connecting them, which is still a common way of visualising social networks. In his book titled *Who Shall Survive?*¹, Moreno analysed the group dynamics of several communities; among them was the Hudson School for Girls in New York, where, during Moreno's field study, a curious incident of 14 runaways happened in a short period of two weeks. By mapping the social network of the students,

¹Even though Moreno is the sole author of this book, it seems that female psychologist Helen Hall Jennings had a considerable role in the data collection and analysis of the studies presented in this book [57].

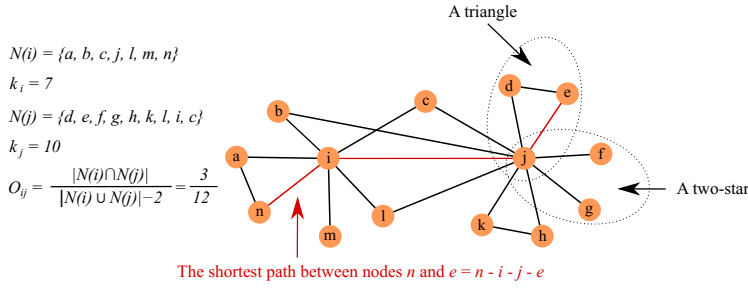


Figure 2.1. Some basic concepts of network science. The neighbourhood of node i is denoted as $N(i)$ which is the set of all the nodes that are adjacent to i (are connected to i with a link). The degree of a node is denoted as k and is equal to the size of the neighbourhood set of the node. The overlap of the link connecting nodes i and j is denoted as O_{ij} and is defined as above. The distance between two nodes is equal to the length of the shortest path connecting them, while the length of a path is equal to the number of links it includes.

Moreno showed that the seemingly independent cases of runaway were part of a chain, and the behaviour was propagating among the students through the social network [110].

Moreno's sociograms were based on data collected through field studies and interviews. Thus, these networks were small and did not exceed dozens of nodes. In contrast, we can nowadays construct social networks consisting of millions of individuals from digital traces. The study of such large-scale networks has considerably advanced our understanding of the properties of human social networks in recent years. In Section 2.4, we review a few of these properties.

Even though the network representation of social relations dates back to 1930's, a structural approach to understanding human behaviour precedes Moreno's work and has been one of the core ideas of modern social science since it was established [21, 62]. I would like to finish this section with a quote from Mark Granovetter. In his keynote at the Sunbelt Social Network Conference in 1990, he tells the story of how a few years after the publication of his famous paper on the strength of weak ties [63], he tried in vain to quit network science:

A funny thing happened on my way out of network analysis: as I innocently pursued these other interests, which I approached from various directions, I always found that as I got more deeply into any subject, network ideas kept coming in the back door. I would write an article that I thought was completely innocent of any network ideas, and someone who had read it would say: oh yes, that's the network approach.

(...)

It shouldn't be any surprise to me that I kept coming back to network analysis, since I am a sociologist, and there really is no way to remain faithful to the fundamental insights of sociology without paying attention to networks of social relationships.

What after all, is the distinctive contribution of the founders of modern sociology, such as Durkheim, Weber and Simmel? It is precisely that one cannot understand social life as the summing up of individuals' motives and traits, as they are given by the study of psychology. This was the fundamental insight of Durkheim in his classic study of suicide – that this most individual of acts is to be explained by the way people are or are not integrated into social networks.

2.1 What are social networks?

The first and maybe the most crucial step in any social network study is defining the network, and this is by no means a trivial task since there are infinite ways to define a social network. The researcher needs to define the network according to the phenomenon she wants to study as well as the type of data she can get her hands on.²

As an example, let's look at the specific case of studying contagion in society. If the incentive is to track the spread of a contagious disease from an infected person to others, then a temporal contact network is ideally needed. Depending on the type of the disease (e.g. airborne diseases, or sexually transmitted diseases), data on different types of contacts are of interest. If, instead, we want to investigate the information contagion, then a wider range of contacts is relevant (calls, online communication, etc.) since the transmission does not require co-location. If the contagion of beliefs is of interest, then besides who is talking to whom, we need to find a proxy for the influentiality of the relationships.

However, in contrast to defining a network consisting of a specific type of contacts or relationships, sometimes the purpose is to capture, in the same network, all kinds of interpersonal relationships ranging from acquaintances to family and romantic partners. The links in such networks are usually weighted, acknowledging that relationships vary in their strength. Building such a network is an attempt to capture a holistic image of the myriad of relationships that connect individuals to each other – ranging from close relationships to which we devote a large portion of our time and which are crucial for our well-being [96] to the most subtle relationships that, while being non-binding, have the potential of leading us to the

²This is, of course, a bit of a chicken and egg problem. Sometimes the topic of the study itself is determined by the available data. Exploratory studies are especially common in data-driven computational social science.

resources out of our everyday social circles.

Reconstructing the holistic social network is required for studying topics that concern individuals' overall social experience, such as loneliness, integration, social support, social capital, influence and status, and beliefs.

2.2 How to detect social ties and measure their strength?

Interpersonal relationships can be inferred from two main categories of data: observational and self-reported. The observational category includes data collected by field studies as well as traces of digital communication and data captured by proximity sensors (such as Bluetooth sensors or RFID tags). In contrast, in the self-reported category, the subjects themselves list their relationships or social interactions (and sometimes the relationships they think exist between people they know).

How to map such data to a social network? How to categorise ties as absent or present? How to quantify the strength of existing ties? Despite decades of social network research, there are no universal solutions to these problems, and in my opinion, never will be. The lack of universal definitions or recipes complicates the matters but is not surprising given the variety of relational data and the intrinsically fluid concept of social ties.

However, there have been attempts to devise *the recipe* for measuring tie strength. For example, in the famous paper *The Strength of Weak Ties*, Granovetter argues:

“Most intuitive notions of the ‘strength’ of an interpersonal tie should be satisfied by the following definition: the strength of a tie is a (probably linear) combination of the amount of time, the emotional intensity, the intimacy (mutual confiding), and the reciprocal services which characterize the tie. Each of these is somewhat independent of the other, though the set is obviously highly intracorrelated.”[63]

In that article, however, Granovetter himself settles for a categorical labelling of ties as absent, weak, or strong and the discussion on how to quantify each of the four factors and combine them into a single measure is postponed to future studies. Here, I will do a (non-comprehensive) review of different approaches to the problem of tie inferences from the literature.

Many social network studies, especially the more traditional ones, rely on data falling under the self-reported category, where the respondents are asked about their relationships and possibly about the relationships they think exist among the people they know. These questions can concern the presence of ties (such as "list all your friends/family members/acquaintances"), or their perceived strengths (such as "rate the emotional intensity of each of the relationships that you have enlisted"), or they may ask about matters which can be indicators of the existence of strong ties (such as

"How often do you and your friend provide each other with emotional support?" used in [118], or "With whom you discuss matters important to you? [147]).

Another common interviewing technique, known as the name generator technique, is to ask the respondent to name a certain category of his social contacts, such as "Whom do you ask for financial advice?" The interviewer may also ask extra questions about attributes of named people, such as the nature of relationships (e.g. family, colleague, friend) or about their socio-demographic attributes (age, gender, occupation, etc.). Such additional inquiries are known as name interpreters [146].

Previously, I discussed the reactivity and recalling biases, which both can affect the reliability of self-reported data. Another challenge for network inference originates from the subjectivity of social relationships resulting in the lack of reciprocity in self-reported data. Two parties of a relationship can have very different perceptions of the strength of the tie between them or disagree on the very existence of it [27, 92]. These discordant perceptions can be combined with deterministic or probabilistic approaches to form a single network [41]. Sometimes it is assumed that a tie is present only if it is mutually acknowledged, in which case the intersection of the reported networks is used (e.g. [88]). Alternatively, the researchers may decide that one nomination is enough evidence for the existence of a tie, thus the union of the reports are used. Similarly, in cases where the reports include some information on tie strength, an average of the reported strengths can be used. Another approach to combining the discordant reports is using probabilistic models. These models may, for example, assign a credibility score to each reporter (e.g. [5]) or assume that different reporters may have a tendency to under-report or over-report their ties (e.g. [41]).

Another approach is to infer ties from people's actions instead of directly asking them to evaluate their ties. This may be done by asking people to keep a diary of their social interactions or by interviewing the subjects about their past interactions. In these cases, the data still falls under the self-reported category. Alternatively, the social interaction data can be collected via field studies (detailed observation of the subjects in their natural environment), this approach falls under the observational category and is not affected by recalling bias. However, the reactivity bias can still affect the quality of the data as people often change their behaviour when being observed.

Social interactions can also be captured in the digital traces that people leave behind. Compared with data collected by field studies, digital traces are less affected by reactivity bias. Moreover, they can include data on the behaviour of millions of people, a scale that is practically not feasible for field studies. All the articles in this thesis rely (fully or partly) on digital traces.

Since digital devices are embedded in our daily lives in numerous ways,

digital trace data comes in all shapes (and indeed sizes). Communication logs are a category often used for inferring social networks. If a person calls another, they are most likely related in some way (with the exception of calling a wrong number or spam calls) and thus a social tie exists between them. The frequencies of calls are typically used as a proxy of tie strengths [116, 139, 108, 163]. This is in line with Granovetter’s hypothesis that the amount of time two people spend together can be an indicator of their tie strength. Saramäki *et al.* [139], in their study including both call logs and some self-reported measures of emotional closeness, showed that there is a high correlation between the perceived closeness and frequency of calls, justifying that call frequency is a good proxy for tie strength.

However, the high correlation does not mean that call frequency is a *perfect* proxy for the tie strength. For instance, Ref. [170] argues that low levels of communication do not necessarily mean that the tie is weak. In their study, many of the participants rarely called or sent a message to people whom they identified as close relations. One possible explanation is that not all relationships require the same amount of communication to maintain their emotional closeness. Family relationships, for example, are known to preserve their emotional closeness even when people do not keep in touch [132, 170]. Moreover, friendships seem to involve more frequent contact in the early phases compared to when they are already established [102]. This may indicate that similar to kinships, old friendships require less maintenance, or as suggested by Wiese *et al.* [170] it may be that the emotional closeness lingers for a while and might decline only after an extended duration of no-contact.

Nevertheless, even if we assume that the amount of time that two people spend together is a good proxy of their tie strength, calling is merely one of the many modes of communication. Monitoring a single communication channel portrays an incomplete and distorted picture of people’s communication behaviour. This is especially true because, as we show in Publication I, people tend to prefer different communication channels for contacting different sets of people. Therefore, a single channel will not provide a representative sample of all communication events of a person. Thus, multichannel studies are essential for conceiving a more complete picture. However, the available data is often far from comprehensive and rarely spans more than a couple of channels for the same population. Another practical challenge is how to quantify tie strength from multichannel data when channels are intrinsically different. For instance, the same conversation that may be concluded over a single call might require dozens of ping-pong text messages. We have addressed this problem in Publication I.

The length of the observation window is another factor that impacts the reliability of networks inferred from observational data. If the window is too short, it fails to capture a considerable fraction of social ties, especially since due to the bursty nature of human communication [89], the inter-

event times³ can be rather long. Krings et al. [89] investigated the impact of the length of the observation window on the properties of the aggregated weighted call network. They monitored the structure of the network while increasing the length of the aggregation window from one day to three months and observed that at first, a collection of disjoint densely knitted communities appeared, and as the window grew beyond a month, weaker links emerged bridging between the communities.

To conclude, I hope you now agree with me that inferring the social network is not a trivial task. The inter-relational data can be biased in many ways. Moreover, when inferring the network from the data, we need to make choices, and they affect the properties of the reconstructed network. Thus, as scientists, when drawing conclusions and interpreting our results, we should keep in mind that the networks we study are just imperfect proxies of the underlying social relationships.

2.3 Shadows of reality

...the truth would be literally nothing but the shadows of the images. [79]

– Plato

I would like to make an analogy between the problem of inferring networks of social relationships and Plato’s cave allegory (as depicted in Figure 2.2). In the previous section, we discussed various ways of inferring the social network from different types of relational data. A core implicit assumption of all social network inference attempts is that there exists a true underlying network (akin to Plato’s Forms). However, we can not directly observe the underlying network. All we can grasp are incomplete and distorted reflections of the underlying network (akin to shadows on the wall of the cave), and maybe the best we can do is to combine these incomplete reflections, an approach taken by multichannel studies (e.g. in Publication I) as well as studies that deal with the problem of inferring the underlying network from a multitude of (often contradicting) reported network data [41].

Having made the analogy, I should clarify how far this parallel goes. Unlike Plato’s Forms, which are timeless and unchanging, the underlying networks are dynamic and constantly changing. Moreover, while the existence of transcendental Forms is the basis of Plato’s philosophy, I think social network analysis can and should conceive a more pluralistic approach to reality: an approach that questions the existence of any *true underlying network*. This is especially important because social rela-

³Inter-event time is the duration between two consecutive communication events of the same tie.

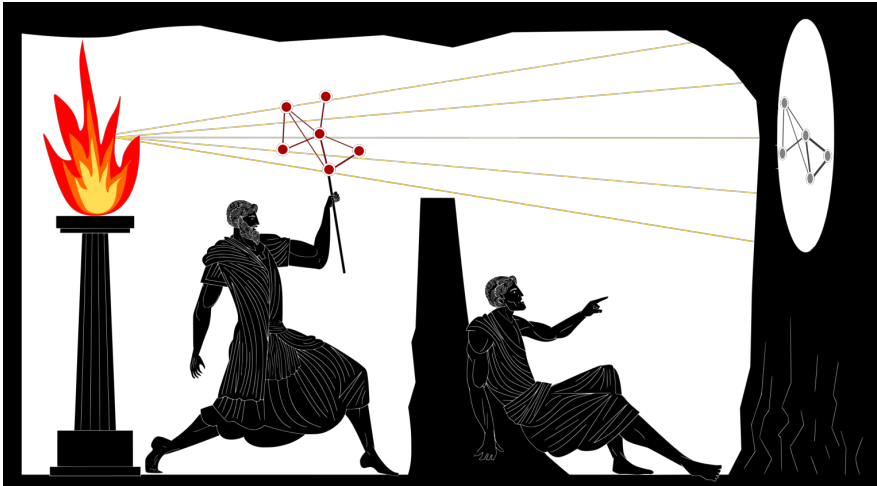


Figure 2.2. An implicit assumption in all attempts to infer social networks is the existence of a true underlying network (similar to Plato's Forms) that we need to recover. Yet, this underlying network remains beyond direct observation. Instead, what we perceive are incomplete and distorted reflections of the underlying network, akin to shadows cast on the cave wall. Copyright note: Some of the elements in the figure are borrowed from an illustration by Matias Del Carmine (account name: matiasdelcarmine) available on Adobe Stock website under file number 255878312. The right to use the picture is purchased from Adobe Stock by the author of the thesis.

tionships are subjective and cognitive concepts [164]. Let's imagine the hypothetical case of asking all the students in a class to map out their perception of the friendships among the students: whom do they consider as their friends? Who do they think are friends with whom? Based on similar studies [109, 92], we expect students to have different perceptions of the social network in the class. None of these subjective perceptions are wrong and they all matter because each social agent strategies their actions according to their own individual perception of the social network [41].

2.4 Characteristics of social networks

Social networks are highly structured and far from random. Decades of studies are devoted to identifying these patterns and explaining them according to human behavioural traits and underlying social processes. Here, I briefly review some of the most important characteristics of social networks from the perspective of network science.

Social networks typically exhibit positively skewed degree⁴ distributions: a few nodes have significantly large degrees while most of the nodes have degrees smaller than the mean [117, 160]. Moreover, social networks show

⁴Degree of a node is defined as the number of its adjacent nodes (its neighbours). Where two nodes are adjacent if there exists an edge between them. See Figure 2.1 for a schematic illustration of degree and neighbourhood.

degree assortativity as the high-degree nodes tend to be connected to other high-degree nodes and lower-degree nodes to each other [112, 113].

The nodes in social networks are locally clustered but nevertheless globally connected with a small average shortest path⁵ length of only a few steps [42, 159, 8]. The global connectivity via short chains is also known as the *small-world* phenomenon, a phenomenon which was empirically observed by Stanley Milgram in his small-world experiments [159] that intended to measure the average *network distance*⁶ between pairs of individuals residing on opposite coasts of the United States [12]. Milgram recruited hundreds of people residing on the West Coast and gave each of them a letter addressed to random recipients living on the East Coast. Each person was instructed to directly send the letter to the recipient if they knew him/her, or otherwise forward it to a relative or friend who was more likely to know the recipient. Milgram reported the average shortest path for successfully delivered letters to be around five and a half.

Several possible underlying mechanisms have been proposed for explaining the simultaneous existence of dense local clusters and a short degree of separation between random node pairs. Granovetter explained this phenomenon by pointing out the different properties of weak and strong ties [63]. He hypothesised that the stronger the tie between two nodes, the larger the proportion of their common neighbours. Thus, strong ties tend to exist in densely knitted communities and are unlikely to function as bridges between the communities. Instead, the bridging ties tend to be weak. He argued that individuals with only a few weak ties are deprived of exposure to novel information (for example information on job opportunities), and on the societal level, political movements can not gain momentum in a society deprived of weak ties [61]. Nevertheless, more recent works indicate that the effective utility of weak ties depends on the socio-demographic attributes of people. Women [23] and minorities [149] benefit less from their existing weak ties [142].

Granovetter's theory [63] was inspired by the social balance theory proposed by Heider [70], suggesting that if two individuals have opposing attitudes toward a third person, their relationship is imbalanced and tensioned. It follows that closed triads are more stable than two-stars, and thus, a more common motif is social networks. In other words, it is more probable that people have a friendly relationship with friends of their friends as opposed to having a hostile relationship that can result in conflicts and psychologically difficult situations.

Recent empirical studies conceived after the turn of the century have utilised digital traces to examine the 20th-century hypotheses of both

⁵The shortest path between two nodes is the path that has the fewest links. See Figure 2.1 for an example.

⁶The distance between two nodes in a network is defined as the number of links in the shortest path between the two nodes.

Heider and Granovetter. Szell *et al.* [155] analysed the network of around 300,000 players of an online game and reported an over-representation of closed triads of positive relationships (predicted by the balance theory [70]) as well as a correlation between communication intensity of ties and their neighbourhood overlap (a property suggested for the strong ties by Granovetter [63]) and a negative correlation between the betweenness centrality⁷ of ties and the neighbourhood overlap of their parties (inline with the bridginess of weak ties suggested by Granovetter [63]). Along the same lines, Onnela *et al.* [117] reported a positive correlation between overlap and weight of links in the weighted mobile phone communication network of 3.9 million users and showed that as Granovetter predicted [63], the removal of weak links disconnects the network faster than the removal of strong links.

Another explanation for the locally dense but nevertheless globally connected structure of social networks is Feld's theory on the focused organisation of social ties [54]. Feld believed that people often organise their social ties not in the void but instead around different foci. A focus is an entity (such as workplaces, schools, hobby groups, or families) that facilitates repeated joined activity among its members and thus fosters tie formation among them.⁸ Foci vary in size and in how constraining they are (for example, family vs neighbourhood). The more constraining foci are more interaction-enforcing and thus create tightly-knit clusters.

Global connectivity, on the other hand, is explained by the involvement of individuals in multiple foci, whether simultaneously or at different periods over time. While a person's contacts (alters) in a focus are likely to be connected to each other directly or through other members, the alters belonging to different foci are less likely to be connected. Thus, the person can act as a connector between different clusters that are each associated with a different focus. People can search and navigate these loosely connected clusters by using their knowledge or hunch about the foci that other people are involved in. This may explain how the participants in Milgram's small-world experiment [159] were able to deliver the letter to the intended recipient via a short chain [165].

Homophily is another mechanism that can result in the local densification of social networks [105]. People tend to form relationships with people who are similar to them [93]. Thus, a person's friends are also likely to be similar to each other and, as a result, have a natural tendency to befriend each other, which can contribute to triadic closure and increase of clustering coefficient [169].

⁷Betweenness centrality of a tie is a measure of the frequency at which a tie in includes in the shortest paths between pairs of nodes in a network.

⁸Feld believed that there is an interdependent relationship between social ties and foci. While ties arise from foci, foci can also arise from social ties. This is because once there exist established ties between people, they tend to find or develop new foci for joint activities [54].

After this brief review of decades of studies of social networks, I will discuss my own PhD work in the following three chapters, hopefully standing on the shoulders of giants.

3. Social egocentric networks and their properties

3.1 What are egocentric networks?

An egocentric network (ego-net) consists of a focal node (ego) and all the ego's immediate neighbours (alters) connected to the ego via social ties (links).¹ These social ties vary in their functionality, emotional closeness, and contact frequency since the set of alters includes a myriad of relations, including family members, friends, acquaintances, and anything in between. To acknowledge this, we can use weighted ego-nets in which a weight is assigned to each link, denoting a measure of tie strength.² See panel (a) in Figure 3.1 for a schematic representation of a weighted ego-net.

3.1.1 Why do we study egocentric networks?

Ego-nets are the smallest units of social networks. Each individual is in the centre of their own ego-net, and these ego-nets join each other like pieces of a jigsaw puzzle and form a larger network (see panel (b) in Figure 3.1 for a schematic illustration). Switching between the ego-nets and the larger network allows the researcher to inquire about her research question at both the individual and population level [39], an approach that we took in Publication IV. Moreover, even though the study of the properties of individual ego-nets cannot reveal the detailed structure of the larger network, we can, for instance, obtain a good estimation of the average degree of the population-level social network by averaging over the number of alters of a large and representative sample of ego-nets [111].

Ego-network analysis is a framework that naturally suits any type of data

¹In some definitions, the ego-net also includes the ties among the ego's alter. However, this is not the case for the definition used in this thesis.

²As discussed in Section 2.2, inferring ties and measuring their strengths are intrinsically challenging, and there are many approaches to this problem.

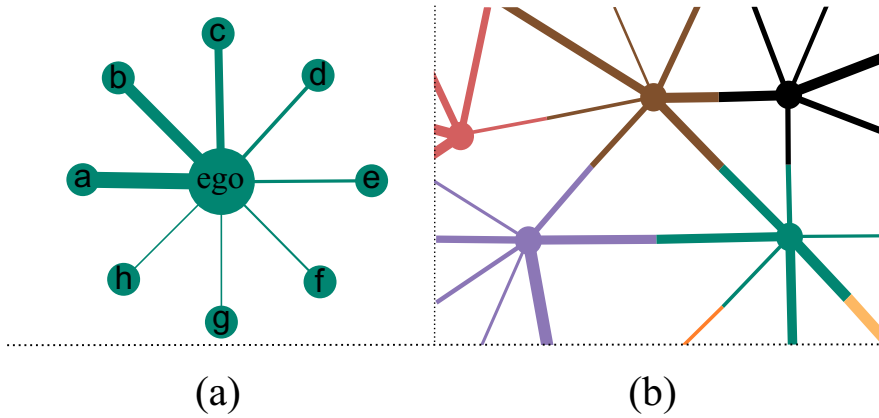


Figure 3.1. (a) A schematic visualisation of an example ego-net, which consists of the ego in the centre connected via weighted links to her alters. The thickness of the lines is proportional to link weights. (b) The ego-nets join each other like pieces of a jigsaw puzzle (each visualised in a different colour) and form the broader network.

in which individuals are the units of inquiry or collection. Such egocentric data collection approaches include traditional surveys (data collected by questionnaires, interviews, or name generators) and many categories of digital trace data, including call detail records (CDRs) datasets.

Egocentric analysis is a subcategory of social network analysis. While a sociocentric approach concerns the totality of social structure, we may take an egocentric approach to focus on an individual and her tapestry of various personal relationships, a tapestry partly created and designed by the ego and whose structure and composition have consequences for her [146]. A multitude of studies suggest that the quality, quantity, stability, and diversity of an individual’s relationships affect her well-being, social status, identity, and access to material and social resources (I elaborate on this in Section 3.1.2).

The structures of ego-nets reflect our shared traits as humans, our individual characteristics, and also the underlying environmental and societal factors. Meanwhile, the network inference process also affects the structure of reconstructed networks (see Section 2.2 for a discussion on the effects of the inference process on the reconstructed social networks). By careful study of the ego-nets, we aim to separate these factors and gain insight into the behavioural traits of people.

3.1.2 Social integration and health

In Section 2.4, I discussed the distinct sociocentric effects of weak and strong ties and how these ties weave us into a highly locally-clustered but nevertheless globally-connected web. Here, I review the importance and

functionalities of weak and strong ties from an egocentric perspective.

Studies show that intimate and strong relationships are crucial for the well-being of different primate species besides humans [74, 73, 171]. When it comes to humans, the essentiality of intimate relationships for well-being, integration, and survival throughout life is undeniable. John Bowlby emphasised the importance of having an intimate and reliable bond already at the stage of infancy [22]. He argued that having a secure attachment with the caregiver is a primary need of the infant just like the need for nutrition and warmth, and the lack of it would result in physiological symptoms of distress. Moreover, he believed that a secure attachment to the primary caregiver promotes the development of self-esteem and may have long-lasting effects on the ability of the individual to form secure and loving relationships in adult life [72, 17]. Furthermore, some studies suggest that not only the existence of close and supportive relationships is essential for the well-being of the individual, but their network arrangement also matters. For example, one study [15] shows that suicidal thoughts are significantly increased in young girls not only by social isolation but also by friendship patterns in which friends of a person do not befriend each other.

While a feeling of safety seems to be associated with having a cohesive network of strong ties, the diversity of weak ties promotes a sense of efficacy [80]. Moreover, being in a network position that provides access to different and separate social circles (or in Burt's terminology, being adjacent to structural holes [25]) gives ideal brokerage possibilities and is associated with higher salaries, promotions, and innovations [24, 80, 26].

Moreover, reviewing the literature on the effects of social relationships on health and well-being suggests that it is not just the mere existence of strong and weak ties in the ego-net that matters but also a perception of being socially engaged and integrated. Studies show that social engagement is a clear predictor of life expectancy [74]. Berkman et al. [17] argue that social engagement, or active enactment of social ties, is a powerful predictor of mortality because through social ties, people practice their social roles (such as familial, occupational, and community roles), and these social roles provide them with a sense of value and belonging and essentially give meaning to life. Some studies particularly suggest that possessing multiple social roles at the same time enhances a sense of self-worth and thus can affect well-being [157, 17]. On the other hand, social isolation and disintegration increase the speed of organ ageing [17] and, therefore, reduce longevity, a phenomenon that might partially explain why immigrants age faster [78, 97].

To conclude, even though we can attribute certain functionalities to strong and weak ties, they both contribute to the social integration of ego. Social integration is fulfilled through the enactment of a combination of these ties. Moreover, as an individual lives through life stages, different

types of ties may become relatively less/more important.

In Publication I, Publication II, and Publication III, we study and model how individuals' simultaneous need for both strong and weak ties form the structural properties of their ego-nets.

3.2 Ego-nets and their structural properties

The interplay of individual characteristics, social environment and norms, as well as our common traits, needs, and limitations as humans determine the structure of ego-nets. Moreover, as explained in Sections 2.2 and 2.3, the inferred networks are inevitably distorted and incomplete reflections of reality, and the choices made by the researchers during the inference procedure affect the properties of the reconstructed networks.

In Publication I, Publication II, and Publication III, we infer the ego-nets from communication logs. An inferred network is reconstructed from the communication events during a certain time window. More specifically, an alter is present in the ego-net if there exists a communication event between the ego and the alter in that window. The weight of the link between the ego and an alter then shows a measure of the volume/frequency of the contacts between the pair of them.

Three basic structural properties of an ego-net are its degree (number of alters), the composition of alters (who these alters are), and how the ego's communication is distributed among the alters (link weight distribution). The study of the dynamics of an ego-net, thus, includes investigating how these three properties (degree of the ego-net, composition of alters, and the link weight distribution) change over time. In Publication I, Publication II, and Publication III, we study these ego-net properties, their evolution over time, and their variations across communication channels and among the population. The findings of these three articles are summarised in this chapter. However, before presenting them, I will review the earlier studies probing these concepts.

3.2.1 The invariability and transformation of personal networks through the ages

We are mortal beings with limited time and cognitive capabilities [106, 95]. While social relationships are essential to us and bring benefits, they also bring obligations and consume a share of our limited time and cognitive resources [108, 148, 156]. As a result, people strategise how to spend their resources on social ties [107] according to their needs, preferences, as well as societal expectations imposed on them. Having limited resources trivially puts a cap on both the number of alters (or ego-net degree) and the amount of resources spent on each of them [133]. Some studies on the

limited size of ego-nets have investigated this phenomenon for a range of primates and reported that the average number of social ties that members of a species can handle is correlated with their average brain size [45].

While studies such as [45] relate the structural properties of ego-nets to the attributes of humans that change on an evolutionary time scale and over generations, another category of studies investigates the effects of recent changes such as modernisation, urbanisation, and novel communication technologies on personal networks.

The 19th-century sociologist Durkheim argued that the rise of individuality in modern society does not undermine the integrity of society. In contrast, the division of labour makes individuals more interdependent [46]: social relationships undergo specialisation, yet society remains intertwined. In 1979, Wellman [166] conducted a study examining the impact of urbanisation on egocentric networks. His findings challenged the notion that urbanisation leads to increased loneliness, suggesting instead that it has "liberated" personal networks from geographical limitations, in the sense that people can maintain social relationships extending far beyond their neighbourhoods [146]. This liberation results in more diverse and specialised personal networks, specialised in the sense that the ego tends to turn to different alters for different kinds of services and support [168].

Along the same lines, numerous studies have explored the impacts of advancements in communication technology on the structure of personal networks. One consequence is the appearance of convenient communication methods that foster person-to-person social interactions (e.g. mobile phone calls and SMS) instead of household-to-household interactions (e.g. visiting your neighbour) or social interactions taking place inside communities (e.g. gathering at the church) [167]. Consequently, communities and communal ways of socialising are declining while person-to-person interactions are becoming more common. Thus the digital-age social networks are less local, tightly-knit, and communal, and more global, sparse, and individualistic. This phenomenon is referred to as "networked individualism" [168].

Several studies suggest that Internet and digital communication methods do not replace in-person communication [66]. Moreover, offline and online communication methods do not form disjoint factions of personal networks, but social ties, especially the closer ones, are often maintained by a combination of online and offline social interactions [36]. Such observations indicate that studying social networks through a single communication channel brings only limited insight and highlights the importance of multichannel studies for gaining a realistic picture of personal networks. In Publication I, we compared the ego-nets maintained via two different communication media, namely calling and texting, and investigated how the egos use these two media in combination to maintain their social ties.

3.3 Ego-net dynamics

Egonets are dynamic. People initiate new relationships while some old relationships fade away [139, 134, 56], prioritise different types of social relationships (e.g. kin, friends, spouses, colleagues) at different life stages [40], cut ties and break up, and lose people to death. In addition to the volatility due to forming new relationships and dropping some others, ego-nets inferred from the communication logs during specific intervals may seem even more dynamic than the ego's social circle because they rarely capture all the existing ties. This is due to the bursty activity patterns of social ties, characterised by short intervals of intense contact interspersed with longer periods of silence [114, 28, 11, 82].

As a result of this fluidity, the composition and size of ego-nets are in constant change.

3.3.1 Ego-net turnover

Researchers have studied the change in the composition of ego-nets through surveys and interviews, as well as by investigating digital traces of communication.

Using surveys and interviews for data collection, Small *et al.* [148] studied the changes in the memberships of the *core discussion networks* (the set of alters that egos turn to for discussing important matters). They surveyed a group of students before and after graduation and concluded that even the core discussion networks, which were believed to be rather stable, undergo remarkable and rapid changes if egos enter new social environments.³

Saramäki *et al.* [139] used ego-nets made from call logs to study the change in the social network of a group of people in a similar setting (graduating from school). They measured the extent of network turnover by using Jaccard index⁴ of the sets of contacted alters during consecutive 6-month intervals. They reported a mean Jaccard index of 0.33 for the top 20 called alters, indicating that call ego-nets are rather fluid in their

³Fischer *et al.* [56] suggest that the large turnover in the membership of an ego-net as a result of a change in ego's context and environment can be explained by Feld's focus theory [54]. I think this is a plausible hypothesis: social ties are formed and maintained in social contexts or foci that encourage/facilitate/enforce communication between their members. When an ego replaces a social focus with another one, the maintenance of the ties with the members of the old focus becomes more difficult, while the new focus provides the ego with new ties, opportunities and responsibilities. See 4.1 for an explanation of Feld's focus theory.

⁴The Jaccard index, a measure of similarity between a pair of finite sets S_1 and S_2 , is defined as $\frac{|S_1 \cap S_2|}{|S_1 \cup S_2|}$, or the ratio of the size of intersection set of S_1 and S_2 to the size of their union set. A Jaccard index equal to 1 indicates maximum stability, while an index equal to zero means that the sets are completely different.

composition.

Some studies suggest that certain categories of social relationships might be more stable than others. For example, a longitudinal survey-based study [50] on the stability of friendships among middle school students suggests that perceived closeness affects the stability of relationships: the friendships categorised as “best friends” by students were observed to be more stable than those categorised as “just friends”.

Miritello *et al.* [107] attributes the level of turnover in egos’ personal networks to their social *strategies*. Some people are *keepers* with a low ratio of added/removed ties to their ego-net sizes while the *explorers* have higher ratios of added and removed ties. They also reported that social strategies are associated with network properties. For example, the clustering coefficient⁵ of a *keeper* is on average twice the clustering coefficient of an *explorer*.

3.4 Social signatures: dynamic, yet persistent

Saramäki *et al.* [139] showed that despite large turnover in the membership of ego-nets, egos tend to preserve their patterns of communication distribution among their alters. The authors coined the term “social signatures” to refer to these persistent patterns. They are called signatures because while the individual patterns show some changes over time, the variation of signatures in the population is considerably larger. Social signatures and their stability are central topics in both Publication I and Publication III. Thus in this section, I provide the mathematical definitions related to these concepts, give a short review of the literature, and explain the gaps that Publication I and Publication III aimed at filling (not necessarily in this order).

3.4.1 Social signatures: the definition

Social signatures are formed based on weighted ego-nets and capture their normalised weight distributions. Thus, for an ego-net in which the weight of the tie between ego e and the i th most contacted alter is denoted as w_{ei} the social signature is defined as:

$$s_e = \left(\frac{w_{e1}}{\sum_{i=1}^k w_{ei}}, \dots, \frac{w_{ek}}{\sum_{i=1}^k w_{ei}} \right), \quad (3.1)$$

which is an array of length k sorted in descending order (k denotes the degree of the ego-net). See Figure 3.2 for a schematic visualisation of an example weighted ego-net and its corresponding signature.

⁵Clustering coefficient of a node is the ratio of the number of existing links between its neighbours to the total possible number of links.

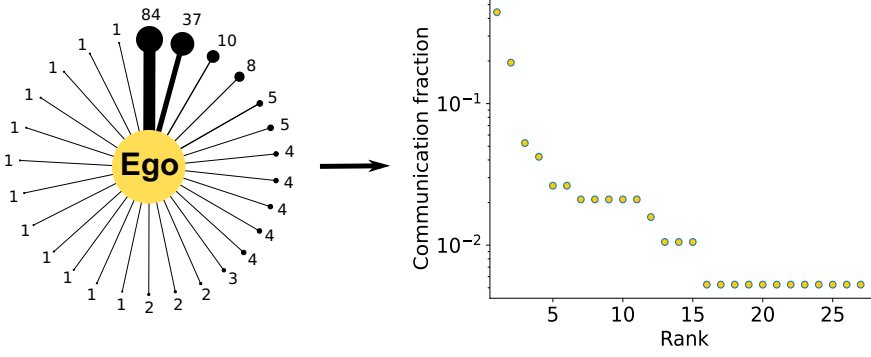


Figure 3.2. An example weighted ego-net (on the left) and its social signature (on the right). Social signature captures the way an ego distributes her communication among her alters. A typical ego-net consists of a few strong ties who receive a disproportionally large fraction of communication while the rest is distributed among a large number of weaker ties.

3.4.2 How to measure the distance between a pair of social signatures?

The concept of signature implies stability and persistence. However, to quantify stability we need to define a measure for assessing the similarity/dissimilarity of signature pairs. In other words, we need to choose a distance function.

In Ref. [139], the difference in shapes of social signatures was measured using Jensen-Shannon Divergence (JSD). In **Publication I** and **Publication III**, we instead used the square root of JSD, which is a distance function.

To measure the distance between two signatures of lengths k_1 and k_2 , we first append a zero-pad of length $|k_1 - k_2|$ to the shorter signature. The JS-distance, d_{JSD} , between social signatures s_1 and s_2 (now of equal length) is defined as

$$d_{\text{JSD}}(s_1, s_2) = \sqrt{H\left(\frac{1}{2}s_1 + \frac{1}{2}s_2\right) - \frac{1}{2}(H(s_1) + H(s_2))}, \quad (3.2)$$

where $H(s_1)$ denotes the Shannon entropy of s_1 and is defined as

$$H(s_1) = - \sum_{r=1}^k f_{1r} \ln f_{1r}, \quad (3.3)$$

with $k = \max(k_1, k_2)$ and $f_{1r} = \frac{w_{ek}}{\sum_{i=1}^k w_{ei}}$ being the r th entry of signature s_1 (fraction of communication dedicated to alter of rank r in signature s_1). For $f = 0$, $f \ln f$ is equated to 0.

Alternatively, the difference in the shapes of signatures can be measured by L^2 distance (or Euclidean distance), which is used in Ref. [139] and also studied in **Publication III**. Similar to the process of measuring JS-distance, to measure the L^2 distance between two signatures of different lengths, we

first zero-pad the shorter signature. Then, the L^2 distance between social signatures s_1 and s_2 , now both of length k , is defined as

$$L^2(s_1, s_2) = \sqrt{\sum_{r=1}^k |f_{1r} - f_{2r}|^2}. \quad (3.4)$$

3.4.3 Distinctiveness and persistence of social signatures

Previous studies have indicated that the shape of an individual's social signature stays remarkably stable over time. This was demonstrated in Ref. [139], where weighted ego-nets were constructed from mobile phone call logs of a group of 24 students, both before and after graduation. The results showed that the shape of ego-nets persisted over time amidst large turnover in their membership.

Ref. [139] took the following approach to simultaneously test the distinctiveness of individual social signatures (from signatures of other individuals) and evaluate their temporal persistence: **1.** The call logs are divided into equal and consecutive time intervals; **2.** A call signature is made for each ego e and interval t , denoted as $s_{e,t}$; **3.** The *self-distance* (the distance between the signatures of an ego in two consecutive intervals) is calculated for each ego. **4.** The *ref-distance* instances are calculated by measuring the distance between all possible pairs of signatures that belong to different people and different intervals ($(s_{e_1,t_1}, s_{e_2,t_2})$ while $e_1 \neq e_2$ and $t_1 \neq t_2$); **5.** The distribution of self-distances and ref-distances are compared. If $\langle d_{\text{self}} \rangle < \langle d_{\text{ref}} \rangle$ and the P -value obtained from Welch's t-test is small enough, the signatures are considered to be, on average, distinctive and persistent, which indeed was the case for the examined signatures in Ref. [139] (figure taken from Publication IV).

In Publication I, we applied the same method for testing the persistence of social signatures using a much larger dataset of more than half a million egos. Our results confirmed the persistence of call social signatures and, this time, with a much larger sample size. Moreover, we examined the persistence of *text signatures* (signatures made from SMS logs) as well as *mixed signature* (made from both call and SMS logs) and showed that they also exhibit the same feature (see Figure 3.3). Note that the social signatures are found to be persistent even though the values of the Jaccard index show considerable turnover in the ego-nets memberships (see Publication I for more details).

However, unlike the ego-nets in Ref. [139], the call ego-nets in Publication I were not made from the number of calls, but by counting the number of *active hours* (See Figure 3.4 for a schematic visualisation of the method). We devised this coarse-grained method for calculating the link weights because it enables us to make comparable call and text ego-nets (you can read more on the comparison of call and text signatures in Section

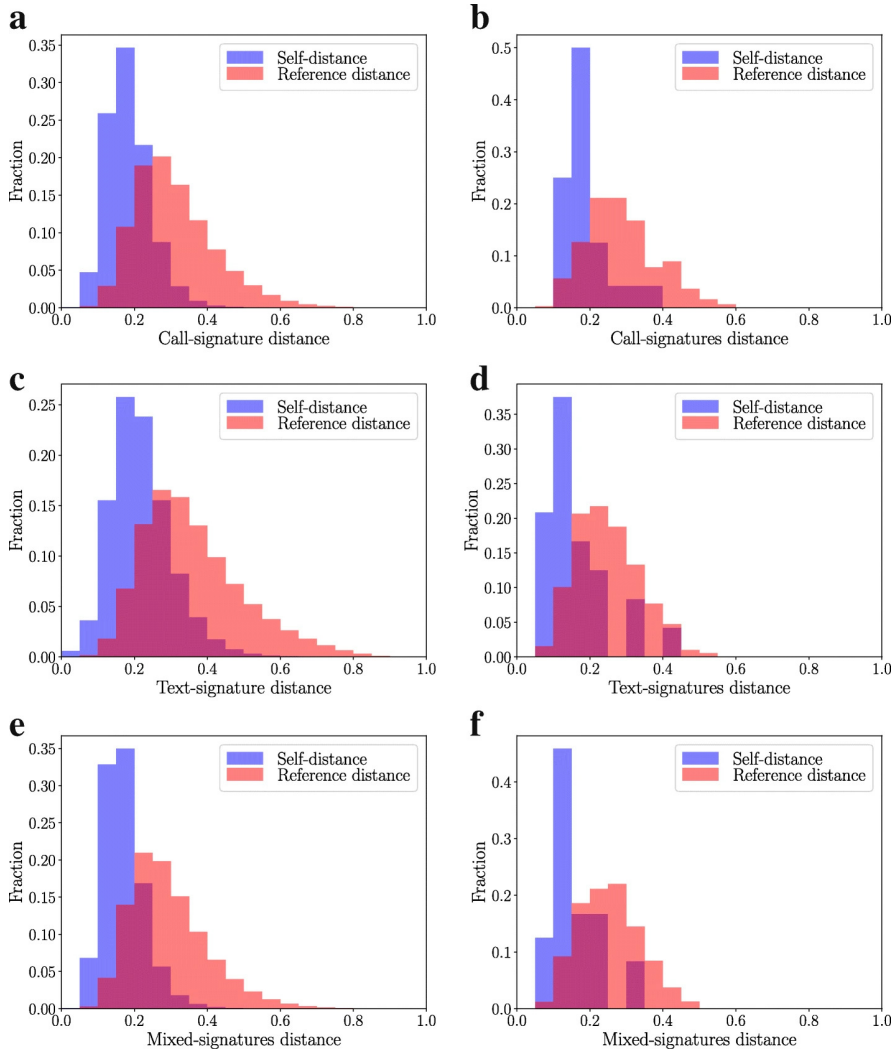


Figure 3.3. Call, text, and mixed social signatures are persistent. In [Publication I](#), we examined the persistence of social signatures made from call logs (plots **a** and **b**), SMS logs (plots **c** and **d**), and their combination (plots **e** and **f**) using two separate datasets. The plots on the left show the persistence of signatures in the larger dataset of approximately half a million egos, while the plots on the right show the signature persistence for the 24 students whose call signatures were also studied in Ref. [139]. In all six plots, the average self-distance is smaller than the average reference distance, which indicates that people typically have persistent signatures. All the distances are measured by the Jensen-Shannon distance. See Sections 3.4.2 and 3.4.3 as well as [Publication I](#) for more details (figure taken from [Publication I](#)).

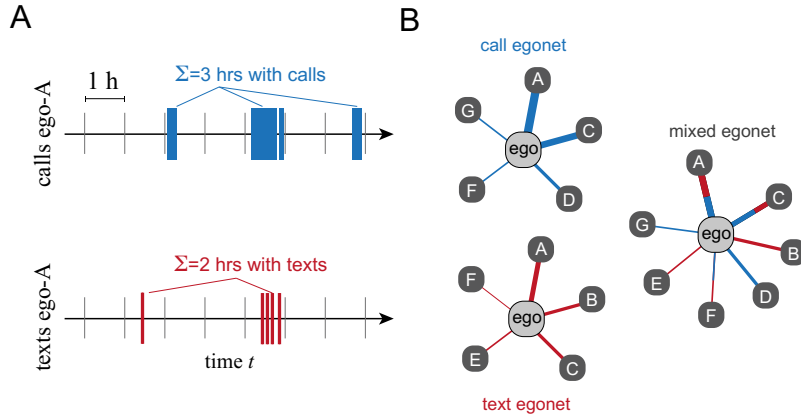


Figure 3.4. Constructing ego-nets from call and SMS communication logs using the *active hours* method for measuring the link weights. Panel A: The timeline is segmented into one-hour time-bins. Then the weight of each tie is determined by counting the number of bins which include at least one event of the desired type(s): call events for call ego-net, SMS for text ego-net, and call or SMS for the mixed ego-net (figure taken from [Publication I](#)).

3.6). Additionally, this method allows us to combine the data on calling and texting to make mixed ego-nets (this method was later employed in Refs. [\[125\]](#) and [\[161\]](#)).

In [Publication I](#), besides providing evidence on the persistence of social signatures, we showed that the call and text signatures of the same individuals resemble each other. This is in spite of the tendency of egos to use these two communication channels for contacting rather different sets of alters. See Section 3.6 for more detailed results.

3.4.4 Persistence, revisited

As explained in the previous section, the persistence of social signatures is commonly examined by comparing the temporal changes of individual signatures with the signature variations in the population (i.e., the distribution of inter-individual distances). If the temporal change of an individual's signature is smaller than the average inter-individual distance, the signature is considered persistent.

In [Publication III](#), we revisited the concept of signature persistence and proposed a method to (a) quantify the extent of persistence for each individual, and (b) disentangle the effects of degree stability and persistence in the patterns of communication distribution.

To quantify the extent of persistence, we utilised the concept of Z-score. The Z-score measures the deviation of a given value (in our case, the temporal change of the signature that we want to quantify its persistence

or $d(s_{t_1}, s_{t_2})$) from the mean of a given reference distribution (in our case, the distribution of inter-individual distances denoted by ref) in units of the standard deviation of the reference distribution. This is expressed mathematically as:

$$Z(d(s_{t_1}, s_{t_2}), \text{ref}) = \frac{d(s_{t_1}, s_{t_2}) - \text{avg}(\text{ref})}{\text{std}(\text{ref})}. \quad (3.5)$$

A negative value indicates persistence as it means that the change in the signature of ego e is less than the average distance between signatures of pairs of different egos in the two intervals. The more negative the value, the more persistent the examined signature is.

Additionally, in Publication III, we investigate the effect of degree stability on the persistence of social signatures. The study that originally introduced the concept of social signatures and reported their persistence, interpreted the stability of signatures as the persistence of “the way that egos divide their communication efforts among alters” [139]. However, such an interpretation overlooks the fact that the shape of a social signature has two aspects: its length (number of alters) and the manner the communication is distributed among the alters (which, in Publication III, we refer to as *tie strength heterogeneity*).

In Publication III, we show that the length difference of a pair of signatures affects their measured distance: the average measured distance is smaller if the signatures have similar lengths. Thus, in order to figure out if signatures are, after all, persistent regarding their weight distribution and if the previously reported phenomenon of persistence of signatures is not solely caused by the stability of ego-nets’ degree, we need to disentangle the effect of the two aspects of social signatures on the measured persistence.

To do so, we calculate the Z-scores (defined by Equation 3.5) with respect to *degree-stratified references*. With this method, the temporal change of an ego with degree change $|\Delta k|$ is compared with a reference consisting exclusively of the ego pairs whose degree difference is equal to $|\Delta k|$.

In Publication III, we also study the $|\Delta k|$ -sensitivity of various distance functions by studying how much the calculated persistence changes when we switch from using a general reference for calculating the Z-score to using $|\Delta k|$ -specific references (more on this in Section 3.5 and in Publication III).

3.5 Disentangling degree and tie strength heterogeneity in egocentric social networks

How does the length difference of a pair of signatures affect their measured distance? Are some distance functions less sensitive to degree difference than others? Can the persistence of social signatures be attributed solely

to the stability of ego-nets' degree? How can we disentangle the persistence of tie strength heterogeneity from the degree stability? These questions are addressed in Publication III.

As discussed in Section 3.1.2, social ties, both strong and weak, are crucial for our well-being. Each person fulfils their social needs and responsibilities and develops a sense of integration into society by weaving their own individual tapestry of social ties.

We would like to examine whether the individual patterns of tie strength heterogeneity are distinctive and persistent because we hypothesise that the distribution of tie strength is fairly stable and may be influenced by individual personality traits and social strategies. To examine this hypothesis, we must compare ego-nets of the same individual over time and across different communication channels. However, as discussed in Section 3.4.4, the shape of the ego-nets has two aspects: the number of alters (degree) and the manner of distribution of communication among the alters (tie strength heterogeneity). Moreover, the distance functions used in the calculation of persistence (and similarity across channels) may be sensitive not only to the difference in the shapes of the ego-nets tie strength distributions but also to ego-nets degree differences. Thus, to be able to examine our hypothesis we first need to disentangle these two aspects.

3.5.1 Measures of ego-net structural distance

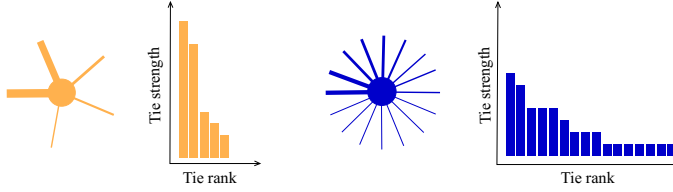
The distance function, used to measure the structural difference of egos, is the cornerstone of concepts such as ego-net persistence or across-channel similarity. Thus, in Publication III we investigate the behaviour of four different distance functions including Jensen-Shannon distance (which is used in Publication I and Ref. [139]) and L^2 distance (see Section 3.4.2). Besides these two signature-based distance functions (meaning that they measure the distance between social signatures), in Publication III, we defined and examined two other distance functions for measuring the structural difference of ego-nets, namely, the *Gini coefficient distance* and the *alter-preferentiality distance* (see Figure 3.5 for an overview of different distance functions examined in Publication III).

Gini Coefficient distance

The Gini coefficient [58] is a widely used metric for assessing the deviation of wealth distribution from ultimate equality, with values spanning from 0 (representing complete equality) to 1 (representing complete inequality). In Publication III, we borrow this notion to quantify the inequality in the distribution of communication among alters of an ego (similar to Ref. [20]).

If the link weights of an ego-net with degree k are arranged in ascending

a) Ego-centric networks and social signatures



b) Measuring structural distances of pairs of ego-centric networks

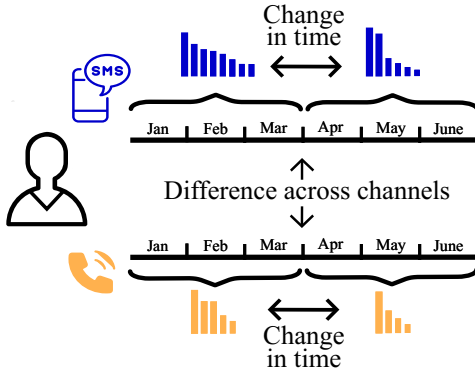
$$L^2 = f(\text{signature}_1, \text{signature}_2)$$

$$\Delta\beta = \frac{|f(\text{signature}_1) - f(\text{signature}_2)|}{f(\text{signature}_1) + f(\text{signature}_2)}$$

$$\text{JSD} = f(\text{signature}_1, \text{signature}_2)$$

$$\Delta G = |f(\text{signature}_1) - f(\text{signature}_2)|$$

c) Intra-individual variation



d) Variation across population



Figure 3.5. A schematic visualisation of the outline of Publication III. (a) Two example weighted egocentric networks along with their corresponding non-normalised social signature. (b) The four distance functions used in Publication III for measuring the structural distance of pairs of ego-nets. The L^2 and the JSD distance functions directly assign a distance value to an input signature pair. In contrast, measuring the distance using the other two functions includes an intermediary step: first, the values of preferentiality parameter β and the Gini coefficient G should be calculated for each ego-net and then the distances can be calculated. (c) and (d) To figure out whether individuals have persistent ego-nets or similar ego-nets across different communication channels, we need to compare intra-individual distances (depicted in (c)) with inter-individual distances (depicted in (d)). The figure is taken from Publication III.

order ($w_1 < w_2 < \dots < w_k$), then the Gini coefficient G is computed as

$$G = \frac{2 \sum_{i=1}^k i w_i}{k \sum_{i=1}^k w_i} - \frac{k+1}{k}. \quad (3.6)$$

The Gini Coefficient distance of egonets e_1 and e_2 with respective Gini coefficients G_1 and G_2 is then defined as

$$\Delta G(G_1, G_2) = |G_1 - G_2|. \quad (3.7)$$

Alter-preferentiality distance

The alter-preferentiality parameter β is a model parameter obtained by fitting our ego-net model developed in Publication II to the communication data of each ego (more about the model and the fitting process in Section 3.7 and in Publication II).

β ranges from 0 to infinity, where $\beta = 0$ corresponds to a fully uniform distribution of communication among alters and infinite β to maximum heterogeneity. Unlike the Gini coefficient, β has no upper bound. Therefore, we propose a normalised distance function which would yield values between zero and one (similar to the Gini coefficient distance). Thus, the distance between a pair of alter-preferentiality parameters, β_1 and β_2 is defined as:

$$\Delta\beta(\beta_1, \beta_2) = \frac{|\beta_1 - \beta_2|}{\beta_1 + \beta_2}. \quad (3.8)$$

3.5.2 Empirical evidence on the importance of disentangling degree and tie strength heterogeneity

In Publication III, we showed that intra-individual degree differences, both degree changes in time and across-channel degree differences (the difference between the degree of call ego-net and the degree of text ego-net of the same ego) tend to be smaller than their inter-individual counterparts (see Figure 3.6). This means that instances of reference-distance distribution are, on average, associated with higher degree differences than the instances of self-distance distributions (self-distance and reference-distances are the distributions that are compared to determine the structural persistent of ego-nets, see Section 3.4.3 for more details).

Considering the above observation, if the measures of ego-net structural difference (i.e. the distance functions) show high sensitivity to degree difference, it is necessary to disentangle the effects of degree before determining whether the level of tie strength heterogeneity is a persistent and distinctive attribute of egos, or in other words, to be able to claim that people have their specific ways of distributing their communication among their alters.

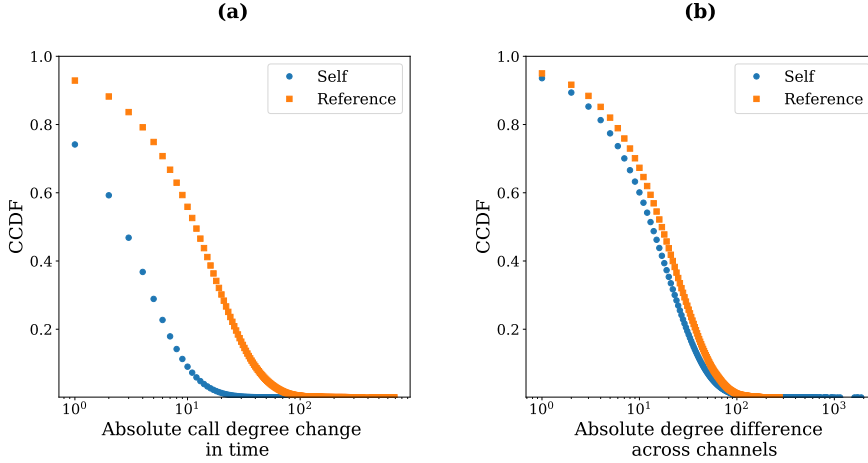


Figure 3.6. Egos have stable degrees in time and to a smaller extent across call and text channels. (a) The complementary cumulative distribution (CCDF) of degree change over time of call ego-nets in our data (blue circles) and the CCDF of the degree differences in the reference distribution (orange squares). The comparison of these two curves indicates that egos exhibit persistent call degrees in comparison to the call degree variation observed in the population. (b) CCDF of the degree differences of call and SMS ego-nets in our data (blue circles) vs. CCDF of the degree difference of the call and SMS ego-nets belonging to random pairs of egos (orange squares). The comparison reveals that egos generally have smaller differences between their call and SMS degrees compared to random pairs. However, this effect is less pronounced compared to plot (a) (figure taken from [Publication III](#)).

In [Publication III](#), we conduct an empirical investigation into the relationship between the measured distance (using each of the four distance functions) and the degree difference of the inputted ego-net pairs. For this purpose, we select a random sample of approximately 7,000 egos and calculate the reference distance for all the possible ego-net pairs that do not belong to the same ego and are not associated with the same time interval (this is similar to how reference distance distributions are formed in both Ref. [\[139\]](#) and [Publication I](#)).

To explore the dependency of the measured distance on the degrees, we group the ego-net pairs based on their degrees in the first and second intervals (k_1 , k_2) and compute the mean distance for each group. Figure [3.7](#) illustrates the average distances for each group as a function of the degree of the ego-nets (in a separate heat map for each distance function). The results indicate that the ego-nets with similar degrees tend to have smaller structural distances. This effect is observed for all four distance functions; however, it is more pronounced for Jensen-Shannon and Euclidean distances and more subtle for the distance functions based on the Gini coefficient and the preferentiality parameter β . Thus, Gini and β based distance functions are more suitable for comparing ego-nets of different lengths.

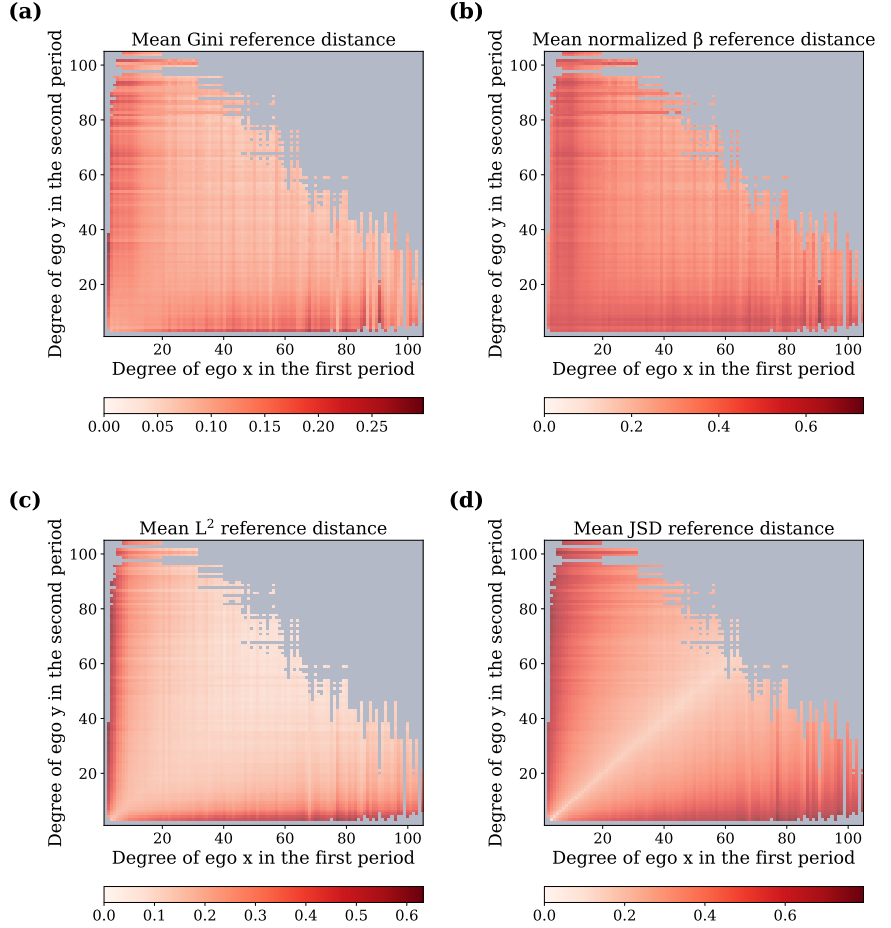


Figure 3.7. Average measured distance is smaller for pairs with similar degrees. The distribution of average reference distances measured by different distance functions [(a) Gini distance, (b) Alter preferentiality distance (normalized β), (c) L^2 distance, and (d) Jensen-Shannon divergence distance]. For all distance functions, we observe that the average reference distance tends to increase as the degree difference becomes larger. However, this trend is more pronounced for the JSD and L^2 distance functions (figure taken from [Publication III](#))

3.5.3 Egos have distinctive and persistent ways of distributing their communication among their alters

Based on the observations explained in the previous section, it is clear that one needs to account for $|\Delta k|$ -sensitivity of the distance functions when measuring the persistence of tie strength heterogeneity ($|\Delta k|$ stands for the absolute value of degree difference). Thus, we use the method of calculating Z-scores with respect to $|\Delta k|$ -specific references (see Section 3.4.4 for instructions on how to calculate the Z-scores).

We computed the persistence Z-scores both against a single degree-agnostic reference, which is a reference with no restrictions on the degree difference of ego-net pairs, [Figure 3.8(a)] and also against $|\Delta k|$ -specific references [Figure 3.8(b)]. As we could predict based on the earlier studies on the persistence of social signatures (e.g. Ref. [139] and Publication I), the Z-scores calculated against the degree-agnostic reference are negative, which means that the social signatures exhibit persistence (regardless of which of the four distance functions is used). Moreover, the majority of Z-scores calculated against $|\Delta k|$ -specific references are also negative. This means that the persistence of social signatures cannot be solely attributed to degree stability and that the tie strength heterogeneity is indeed persistent. People are, after all, distinctive and persistent in how they distribute their communication among their alters.

Moreover, we noticed that when switching to Δk -specific references, the persistence values obtained using the L^2 and JSD distances change more drastically than the values based on β and Gini [see the black and orange pairs of scatterplots in Figures 3.8 (c) – (d)]. This observation suggests that the distance function based on the inequality measure G and the alter preferentiality β are more appropriate for comparing ego-nets with varying sizes, as they are relatively less sensitive.

3.6 Comparing call and text ego-nets

We shape and maintain our personal social networks through various communication channels, such as face-to-face interactions, phone calls, SMS, emails, and a diversity of social media. These channels differ in their functionality and capabilities, leading individuals to use them selectively rather than interchangeably. Several factors influence an ego's choice of channel for communicating with each alter, such as the accessibility of the channel for both ego and alter (e.g. having a phone, knowing how to use a certain social media, the geographically distance with the alter), the type of relationship (e.g. family, friend, colleague), the timing of the communication (people tend to text instead of calling in the late hours), and the purpose of the communication. Thus, different communication

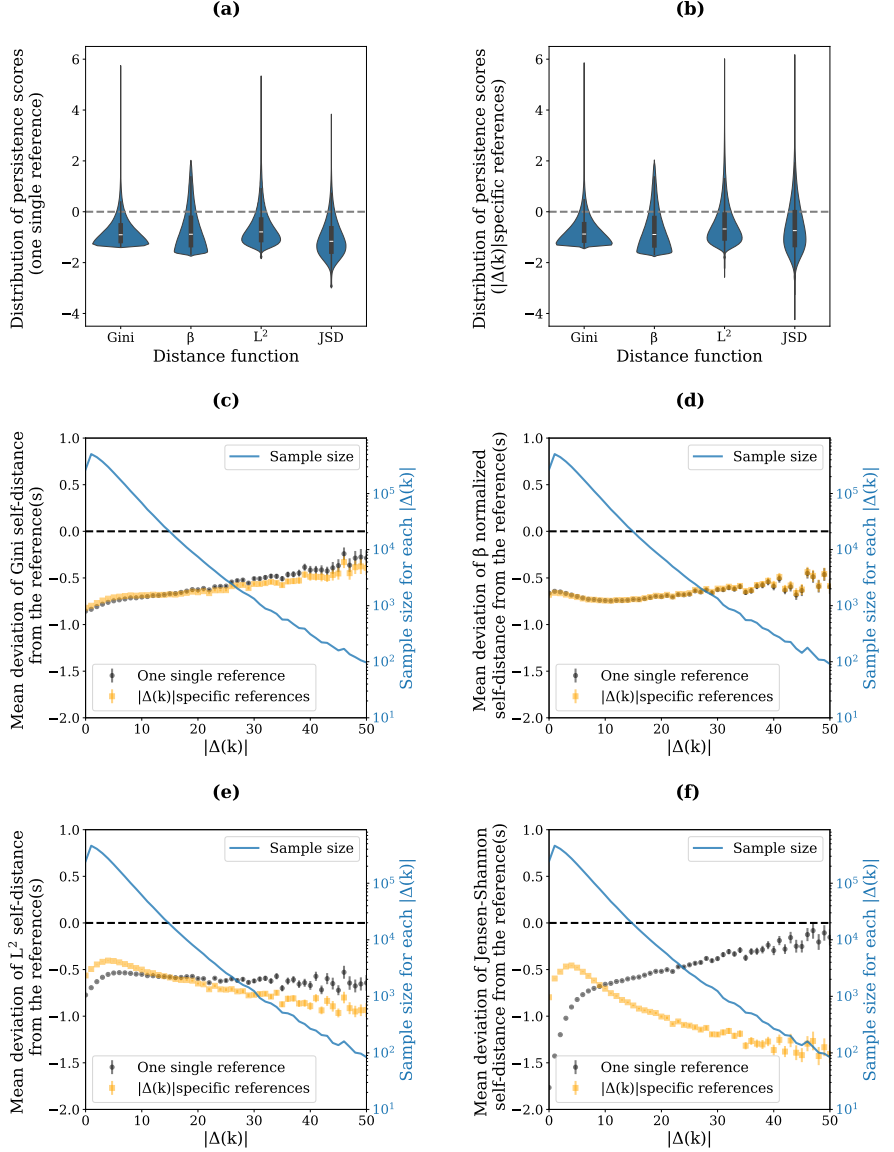


Figure 3.8. Structural persistence of call ego-nets in our dataset using different functions for measuring structural distance. (a) Distribution of persistence scores when measured against a degree-agnostic reference and in (b) when measured against $|\Delta(k)|$ -specific references. In both cases, most of the scores are negative, which indicates the stability of call ego-nets. Average persistence as a function of $|\Delta(k)|$ (absolute degree change) is shown in (c) using the Gini distance function; in (d) using the normalised β distance; in (e) using the L^2 distance; and in (f) using the JSD distance. The persistence values obtained by using all four distance functions have negative average values across the degree range. Moreover, the persistence plots for the Gini coefficient and β do not change much when switching from degree-agnostic to $|\Delta(k)|$ -specific references. This suggests that Gini and β are better measures for comparing ego-nets of different sizes. The blue curves in plots (a), (b), (c), and (d) show the number of egos with a particular absolute degree change $|\Delta(k)|$ (figure taken from [Publication III](#))

channels may give very different pictures of the ego’s personal network. Consequently, when constructing ego-nets, incorporating multiple communication channels provides a more realistic portrayal of the underlying personal network.

However, data spanning several communication channels of the same egos is rare. Moreover, even if we can get our hands on such data, the process of inferring a single network from it is not trivial. For instance, how many text messages count for an hour of being together in the same location?

In Publication I, we took a multichannel approach to study ego-nets based on a dataset including mobile phone call and text message (SMS) logs of approximately half a million egos. As explained in Section 3.4.3, we constructed call, text, and mixed ego-nets for each ego (where mixed ego-net is inferred from both calling and texting data) according to the method of active hours (visualised in Figure 3.4). As explained before, we showed in Publication I and Publication III that the structure of these ego-nets is distinctive and persistent in time.

In addition to studying the persistence of these ego-nets, we applied the very same method of comparing self-distances with reference distances to compare the structural similarity of the call and text networks of the same egos. In Publication I, we observed that egos’ call and text signatures resemble each other, and in Publication III, we showed that the previously observed similarity holds even when taking into account the degree-sensitivity of the employed distance functions.

Moreover, in Publication III, we used the two measures of tie strength heterogeneity, namely, the Gini coefficient G and the preferentiality parameter β to investigate if the call and text ego-nets of an individual are correlated in terms of tie strength heterogeneity. The values showed a moderate correlation for both Gini and β . This is another evidence that people tend to shape their ego-nets associated with different channels in a similar way.

In Publication I, we also studied the relation between the choice of channel (call or SMS) and the ranks of alters in the mixed signature. We observed that top-ranked alters, especially the first and the second strongest ties, are more likely to be contacted by a combination of both call and SMS. This result is in line with earlier studies reporting that media multiplexity (using several channels for maintaining a tie) is in general indicative of a stronger tie [69]. Quadri *et al.* [125] also reported similar results for the specific case of combining call and SMS communication using a different mobile phone dataset.

3.7 An ego-net growth model

In Publication II, we aimed to provide an explanation for the stable and distinctive structure of ego-nets or, in other words, for the existence of social signatures. We did so by making an ego-net growth model that connects people's communication strategies to the particular shape of their ego-nets.

We showed that the varying levels of heterogeneity manifested in the ego-nets can be modelled by a simple mathematical model (inspired by Price's model [122, 123]) that attributes the observed variation to two competing mechanisms of tie reinforcement: (i) cumulative advantage, where ties that are already strong get even stronger, and (ii) random choice of communication partners. A tunable parameter α determines how much the ego is inclined to contact the people she already knows and has an existing relationship with, as opposed to contacting people regardless of the relationship history. In this model, the ego chooses to contact one alter at each time step, consequently increasing the alter's activity by one. The probability of contacting a specific alter with activity a at time τ is then

$$\pi_a(\tau) = \frac{\alpha + a}{\sum_{j=0}^k (a_j(\tau) + \alpha)}, \quad (3.9)$$

where k is the total number of candidate alters for communication, and α is the tuning parameter of the model: a personal attribute of the ego, which determines to what extent the ego chooses the alters randomly or according to their activity scores. When the parameter α is small, π_a increases linearly with activity, so ego interacts preferentially with the most active alters, while for large α , alters are chosen uniformly at random.

Then, we treat our model analytically by solving a master equation for p_a (probability that a randomly chosen alter has activity a) in the limit $\tau, k \rightarrow \infty$, which enables us to explicitly write a maximum likelihood estimation for the *preferentiality parameter* β ,

$$\beta = \frac{\tau/k - a_0}{\alpha + a_0}, \quad (3.10)$$

where α is the previously discussed tuning parameter and a_0 is the minimum alter activity in the ego-net. Then using the maximum likelihood method, we determine the β parameter for each ego-net by fitting our model to the empirical data and then attribute a goodness-of-fit value to each β via the standard Kolmogorov-Smirnov statistic test [152] (see Publication II for more details).

The preferentiality parameter β reveals a crossover in the behaviour of the model where $\beta > 1$ corresponds to the heterogenous regime (in which the alter activity distribution looks like a Poisson distribution), $\beta < 1$ to the homogenous regime (in which the alter activity distribution looks like a gamma distribution), and $\beta = 1$ is the crossover threshold. We observed

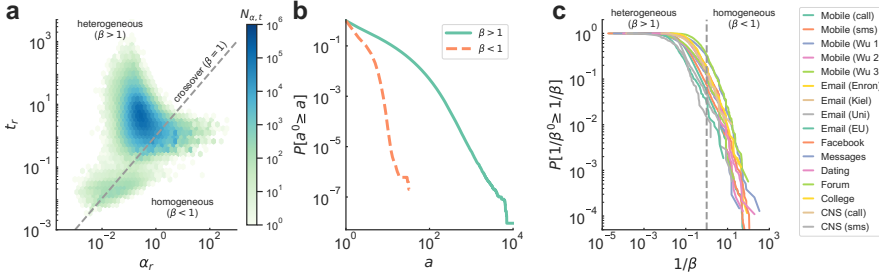


Figure 3.9. Fitting our model to real-world communication data reveals the diversity of social signatures. (a) Heatmap depicting number of egos with given values of $\alpha_r = \alpha + a_0$ and $t_r = \tau/k - a_0$. Most egos (95%) in the studied mobile phone call dataset fall above the $\beta = 1$ crossover line and have heterogeneous social signatures, while a small percentage of the egos (5 %) fall into the homogenous regime and have more flat signatures. (b) CCDF showing the fraction of alters with activity of at least a ($P[a' \geq a]$), aggregated over all ego-nets included in the subfigure (a) that fall into the heterogeneous regime (the green line) or homogeneous regime (the dashed orange line). (c) CCDF of $1/\beta$ fitted for 6.57 million ego-nets in 16 communication datasets (see [Publication II](#) for a description of all datasets). Regardless of the communication channel, the heterogeneity distributions of signatures follow a similar trend, with the majority of ego-nets (66-99 %) falling into the heterogeneous regime and a small percentage of ego-nets (1-34 %) falling into the homogeneous regime (figure taken from [Publication II](#)).

that most of the egos ($\sim 95\%$) fall into the heterogenous regime (see [Figure 3.9](#)).

The goodness-of-fit test allows us to test if our model is appropriate for describing the data (by checking that we can not reject the hypothesis that the empirical data is drawn from the model activity distribution). For the case of the large mobile phone dataset, which we also studied in [Publication I](#) and [Publication III](#), 70% of call ego-nets and 63% of SMS ego-nets pass the goodness-of-fit test and can be attributed a valid β .

As explained in the previous section, in [Publication III](#), we used a distance function based on β to study the persistence and across-channel similarity of ego-nets. Moreover, we did a detailed comparison between the preferentiality parameter β and the Gini coefficient and showed that while these two measures are empirically correlated and theoretically closely related, each of them has its pros and cons.

3.8 Discussion

In [Publication I](#), [Publication II](#), and [Publication III](#) we studied the structure of egocentric networks inferred from a large variety of digital trace data to gain insight into how people organise and maintain their personal networks. We observed that ego-nets typically consist of a small number of close ties which receive a disproportionally large share of egos' social time while the remaining time is distributed among a larger number of

weak ties. The heterogeneous distribution of communication among ties seems to be a universal attribute of personal networks that holds true for most of the ego-nets regardless of communication channel. However, while ego-nets are commonly heterogeneous, there is substantial variation in their extent of heterogeneity. The results of Publications I–III consistently point out that people have distinctive and persistent ways of distributing their communications among their alters. While everyone has limited communication resources, and the majority of these resources are spent on a handful of strong ties, some people tend to invest more heavily in their closest ties, while others prefer to have larger ego-nets, typically at the expense of spending a smaller fraction of their time with their closest ties. Moreover, even though ego-nets exhibit large network turnovers over time and the set of alters that people socialise with are in constant change, egos preserve their individual patterns of communication allocation.

In Publication I and Publication III, we compared the structure of call and SMS ego-nets and observed that the call and SMS ego-nets of a person commonly resemble each other. This is in spite of the fact that typically these two channels are used for communicating with sets of people that do not overlap much.

Why do the shapes of our ego-nets persist in time, even when the people with whom we are in contact change? According to the continuity theory in gerontology, people are inclined to maintain their sense of identity by staying consistent in their preferences and by engaging in familiar roles and situations. Thus, people tend to re-engage in similar social connections if they lose an existing one [56]. Behavioural patterns such as serial monogamy, and friendship styles [131] are among the evidences backing up this hypothesis.

Another possible explanation is Dunbar's theory on the layered organisation of personal networks [175], claiming that people organise their relationships in layers which are associated with different functionalities and varying levels of intimacy. People can accommodate a limited number of alters in each layer regardless of who these alters are.

The stability in the shapes of individual ego-nets and their variation across the population also suggests that they may reflect the personality traits of individuals as these traits also exhibit stability at the personal level, yet considerable variation across the population. Centellegher *et al.* [32] showed that some personality traits are associated with the stability of social signatures, but did not find any relation between the specific shapes of social signatures of people and their personality traits. However, that study was done on a small data consisting of 93 individuals. Future studies on larger samples are needed to draw conclusions.

In Publication II we made a model based on the assumption that while everyone tends to socialise mainly with people that they know well enough, people have varying levels of preferences in their tendency to hang out with

familiar faces or, in other words, to reinforce their ties via a cumulative advantage process. These personal tendencies are captured with the preferentiality parameter β of our model which is determined by fitting the model into the communication data of each ego. In Publication II, we observed that the preferentiality parameter remains stable in time, which may indicate that with this parameter, we are capturing a stable behavioural trait. Additionally, in Publication III, we gained more evidence in favour of this hypothesis by observing that the β values calculated for call and SMS ego-nets of the same egos are correlated.

However, does having personal and persistent communication styles, or in other words having a social signature, explain the structural similarity of a person's call and text ego-nets? I would say, yes and no. People maintain their personal social networks via numerous communication channels (face-to-face interaction, calling, texting, posting on social media, etc.) and thus, as explained in Section 2.3, the ego-net constructed from the communication via a single channel gives only an incomplete and distorted reflection of the underlying personal network. However, since all these channel-specific networks are different reflections of the same underlying personal network, they are bound to somewhat resemble each other in shape unless the reflections are so distorted that the underlying network is not recognisable.⁶ This would, for example, happen if a communication channel includes a negligible portion of an ego's communication events (To avoid this risk, in Publication I, we only studied the egos that were actively using both call and SMS channels).

It should also be noted that our observation of the resemblance of the structure of ego-nets across the channels was limited to comparing call and SMS ego-nets. Further study is needed to determine if this is the case for other pairs of communication channels.

Our results in Publication I, Publication II, and Publication III are indicative of the structural persistence of ego-nets over periods ranging from a few months to a couple of years. Further studies on extended longitudinal data are needed to determine if social signatures remain persistent e.g. during a decade or during the life course.

Moreover, the distance functions used for measuring ego-nets' structural distance are the cornerstone of all the conclusions drawn about persistence and across-channel similarity of the ego-nets. Even though the main focus of Publication III was on the properties of these distance functions (such as their degree-sensitivity), further studies are needed to fully understand them. What are the structural distance functions exactly measuring? For example, is having a stable level of communication with the first top

⁶I would like to make a parallel to Plato's theory of forms and his cave allegory to explain this: the same way that based on Plato's theory all the *particulars* in the material world are merely distorted reflections of underlying perfect forms, still when seeing two particular horses (which are both distorted reflections) one can distinguish that they are both horses.

alters (e.g. a spouse and a best friend) enough for the persistence of the signature?

4. Temporal Multiplexity

4.1 Tie multiplexity

Measuring communication intensity (whether in volume or in frequency) is not the only way to distinguish between weak and strong ties. Social ties are formed and maintained in social contexts. *Multiplexity*, or the existence of multiple contexts in the same relationship, can be an indicator of a strong tie [63]. Multiplexity and high communication intensity usually co-occur (if two people share several contexts, they probably spend more time together, and if two people spend an extended amount of time, most likely more than one context underlies their relationship). However, spending time together in several contexts can be a more effective way of getting intimate than spending the same amount of time together but in one single context. Goffman explained in *The Presentation of Self in Everyday Life* [59] that people actively give different impressions of themselves in different contexts or in each of their many social roles. Some of these impressions may contradict each other. Thus, each individual (the *performer* in Goffman's terminology) tends to segregate his *audience* so that the alters who usually witness him in one role will not witness him in a contradictory role. The segregation of the audience protects the credibility of the impression that the performer wishes to present of himself and prevents uncomfortable situations for both the performer and the audience. I argue, then, that sharing several contexts in the same relationship is an act of confiding, which can be a sign of a close and safe relationship.

Moreover, according to Feld's theory on the focused organisation of social ties, the multiplexity of a tie can affect its functional properties [54]. Feld argued that social ties are often associated with social *foci* – psychological, legal, or physical entities around which social activities happen. He proposed that the more foci underlie the tie between two people, the larger their alter overlap, and thus, the tie is less bridging. In contrast, monoplex ties and ties that are not based upon any focus are more likely to connect

separate clusters. Thus, focus theory suggests that identifying the foci underlying the ties can be the key to understanding the structure of social networks [55].

However, identifying the underlying foci of social ties usually requires careful field observations or data collection methods such as interviews. These methods are not only prone to a variety of biases but also prohibitively expensive and thus not scalable (see Chapter 1 for a review of these biases). Moreover, many of the large-scale social network studies are based on digital traces, such as call detail records, which lack any information on the underlying foci of each interaction. To mitigate this problem, in Publication IV, we investigated the viability of inferring the underlying foci from the temporal manifestation of communication events in the absence of any background information on the social contexts associated with the social ties.

4.2 Can we infer the multiplexity of a social tie from its temporal manifestation?

Digital trace data, such as call detail records, often lack in-depth information about the nature of the social ties. However, they are rich in the sense that they capture the temporal manifestation of social relationships with a high resolution of seconds. Previous studies have shown that the temporal expressions of social ties reflect the nature and context of the relationships [47, 4, 7], their stability [162], and the socio-demographic attributes of the involved parties [87]. Temporal expressions of social ties are reflective of their nature partly due to the existence of cultural norms that regulate the temporality of human behaviour [47]. These norms govern the daily, weekly, and annual rhythms of the time spent at work, at hobbies, with family members, or with friends.

In Publication IV, we studied social ties from an unprecedented angle, namely inferring monoplexity/multiplexity from the temporal rhythms of ties. We proposed a method for identifying different categories of *social times* from the population-wide data as a proxy of different categories of social contexts and then quantified the extent of *temporal multiplexity* of each tie based on its usage of the identified social times. We examined the validity of this approach for detecting monoplex/multiplex ties by comparing the functional properties of the categorised ties. We analysed our results in the framework of Feld’s theory of the focused organisation of social ties [54].

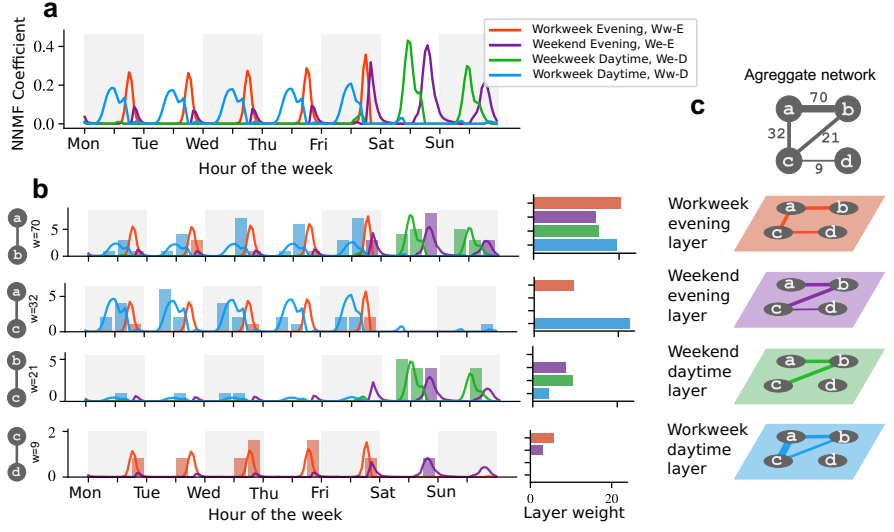


Figure 4.1. Identifying the latent social times and constructing the multilayer network. (a) The four latent signals (social times) extracted from the population activity matrix. (b) Each tie has its own weekly rhythm (The activities of four example ties are demonstrated on the left), we deconstruct these rhythms in terms of the social time signals and obtain the layer weights for each tie (the right-side plots). (c) Using the layer weights we can construct a multilayer network where each layer corresponds to one of the social times (figure taken from [Publication IV](#)).

4.2.1 Data

In [Publication IV](#), we used a large CDRs dataset including metadata on mobile phone calls of ~ 6.5 million users in a European country during the first seven months of the year 2017. This dataset has been studied in [Publication I](#), [Publication II](#), and [Publication III](#) albeit from different angles and with different preprocessing procedures. In the preprocessing procedure for [Publication IV](#), we imposed a reciprocity condition to filter out unidirectional ties that originate from large-degree users, which are most likely marketing and advertisement phone lines. Excluding these ties is crucial because they largely affect network connectivity [[117](#), [116](#)], which is among the topics examined in [Publication IV](#). Moreover, we excluded the ties which had only one call during the whole seven months. We studied the tie multiplexity only for the ties where both parties are the users of the operator company. However, for overlap analysis, we additionally included non-company users (demonstrated in [Figure 4.3](#)). This left us with ~ 10 million ties.

4.2.2 Weekly profiles and extracting the latent social times

To study the temporal multiplexity, we started with forming a *weekly profile* for each tie. A tie's weekly profile is a vector of length $24 \times 7 = 168$,

consisting of the total counts of the tie’s calls (during the seven-month span of the data) placed in each hour of the week (the idea of weekly profiles is similar to the daily profiles defined by Aledavood *et al* [4]). We defined the population weekly activity matrix X as an $N \times 168$ matrix including the weekly profiles of all ties.

We applied Non-Negative Matrix Factorization (NNMF) with an orthogonality constraint [43, 84] on the population activity matrix to cluster the hours of the week into four categories. We also tried clustering the hours into three, five, and eight categories, however, we decided to use four categories because it brought the most interpretable results. See supplementary materials of Publication IV for more details. NNMF is a clustering method commonly used for the dimensionality reduction of temporal data (e.g. [2, 31]). Essentially, by using NNMF, we can approximate the large and usually sparse non-negative matrix X (of size $500,000 \times 168$ in our case) as a multiplication of two smaller non-negative matrices G and H (of sizes $500,000 \times 4$ and 4×168 respectively). The four rows of matrix H are the latent social time signals where each entry shows the strength of the respective signal at a certain hour (see Figure 4.1 for a visualisation of social times). The rows of matrix G include information on how the ties divide their communication events across the four latent social times. The orthogonality constraint promotes independence between the rows of H or, in other words, the distinctness of the social times and facilitates their interpretability. We named each of the social time signals based on the time of the week when they are more prominent, respectively: Workweek Daytime (Ww-D), Workweek Evening (Ww-E), Weekend Daytime (We-D), Weekend Evening (We-E).

4.2.3 Layer coefficients and construction of the multilayer network

Having obtained the latent social times from the population activity matrix, we set out to express each tie in terms of these social times or, in other words, to determine the *layer coefficients* for each tie (we refer to them as *layer coefficients* because, later on, we used them to construct a multilayer network in which each layer corresponds to one of the social times). As mentioned previously, the rows of matrix G already include information on how the ties divide their communication events across the four latent social times. These values can be used to express each tie as a linear combination of the latent social times. However, the calculated matrix G has many near-zero elements, which makes the interpretability of the results difficult and may result in overfitting. It would be easier to deal with a sparse matrix which has zero values instead of many near-zero values. Thus, instead of directly using rows of matrix G for explaining the ties in terms of the social times, we used a statistical inference method for

obtaining the layer coefficients. This method involves a model selection for decomposing each tie into the social times. By selecting the models, we determined which social times are active (non-zero coefficients) and which are non-active (zero coefficients) for each tie. Since we have four social time signals, there are 15 activity arrangements to be chosen from (1 arrangement with all signals being active, 4 with one active signal, 6 with two active signals, and 4 with three active signals). See Publication IV and its supplementary materials for a detailed description of the statistical inference method used for determining the layer coefficients of ties.

Eventually, for each tie i , we formed the layer coefficients vector α_i that sums to one. We used these probability vectors to decompose the ties' weights into four layers, each corresponding to one of the social times, that in turn were used for constructing the multilayer network. The weight vector of tie i with total weight w_i is then calculated as $\hat{w}_i = w_i \alpha_i$ and determines the layer-specific weights of the tie. While almost all the nodes are present in all the layers of the reconstructed multilayer network, a tie may be present in one, some, or all the layers. See Figure 4.1 for a schematic visualisation of the pipeline of constructing the multilayer network.

The reconstructed multilayer network demonstrates a balanced distribution of ties across the layers. All layers contain $\sim 55 - 70\%$ of the social ties, with the weekend-daytime layer being the sparsest and the weekday-daytime layer being the densest. We observed that 19.1% of ties are monoplex (are present in only a single layer). Out of the 19.1% share of the monoplex ties, 2.2% of them belong to the weekend-daytime layer, and the other three layers each include $\sim 5 - 6\%$ of the monoplex ties. In Figure 4.2 (a) you can see the distribution of monoplex, duplex, triplex, and quadruplex ties across the range of total tie activity (w_i).

Eventually, we quantified the *temporal multiplexity* of a tie as the Shannon entropy of the distribution of its layer coefficients $S(\alpha) = -\sum_{l=1}^J \alpha_l \log(\alpha_l)$ (a concept also used for measuring the heterogeneity of social signatures in Publication I and Publication III). Shannon entropy gives a measure of information or uncertainty in a probability distribution. The entropy of a monoplex or mono-layer tie is zero because all contacts can be accounted for by that layer alone, resulting in no uncertainty associated with the layer allocations. The entropy is maximised when all layer coefficients are equal.

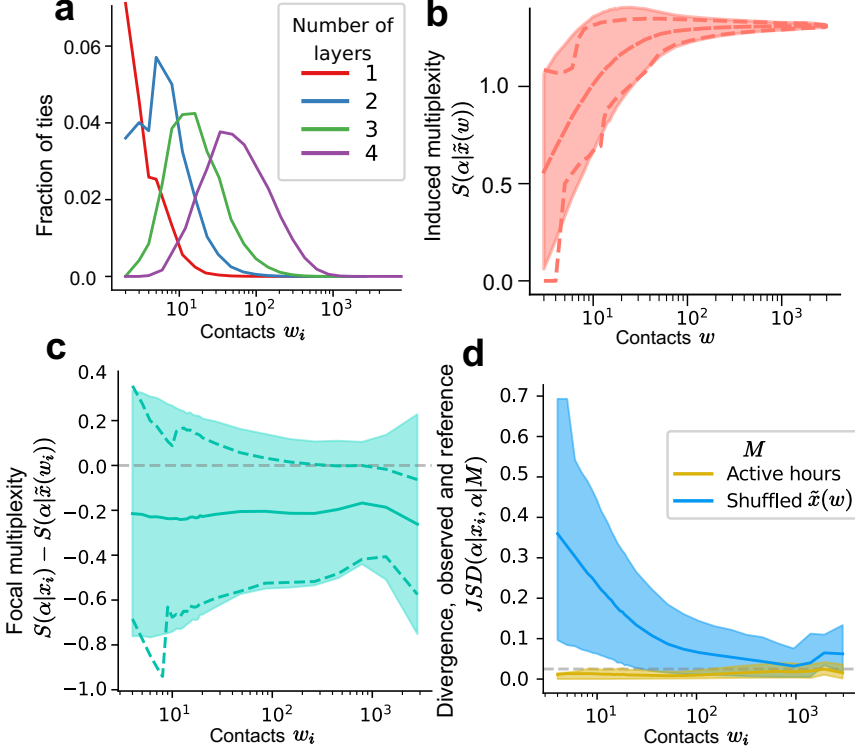


Figure 4.2. Tie multiplexity and the effects of total tie activity and burstiness. We observe that high-activity alters tend to be present in more layers (panel (a)). To untangle this trivial source of multiplexity (namely, high contact frequency) from the tie-specific tendencies, we define *induced multiplexity* (panel (b)) and *focal multiplexity* (panel (c)). Average induced multiplexity increases with total activity w , indicating that multiplexity is expected to increase by activity. However, the average focal activity is negative across the range of w , indicating that ties tend to favour their own social times and this holds across the activity range. (d) We use Jensen-Shannon Divergence (JSD) to compare the layer coefficients of tie call sequences with the obtained layer coefficients when using active hours instead of the number of calls (orange curve). Using the active house dampens the potential effects of burstiness on tie multiplexity. As a reference, we calculate the JSD between the layer coefficient of the ties and the layer coefficient obtained from shuffled sequences (blue). We observe that the layer coefficients do not change as a consequence of applying the active-hours method (figure taken from [Publication IV](#)).

4.3 Results

4.3.1 Ties favour their own social times

As discussed in Section 4.1, multiplexity and high communication intensity often co-occur. Our results show that this statement is also true when it comes to temporal multiplexity: The higher-activity ties tend to be active during more social times and consequently be present in more layers of the reconstructed multilayer network (as visible in Figure 4.2 (a)). In Publication IV, we introduced the concepts of *induced multiplexity* and *focal multiplexity* to untangle a trivial source of multiplexity (namely, high contact frequency) from the tie-specific tendencies.

The expected induced multiplexity of a tie with contact frequency w is determined by producing a random ensemble of ties with the same contact frequency w and averaging over their temporal multiplexities. The ensemble consists of ties with activity frequency w , and the timings of contacts are randomly drawn in accordance with the population's aggregated weekly rhythm. In other words, if, for example, 2% of the population communication events are placed on Tuesdays between 10 and 11, in a shuffled instance, each of the w contacts is allocated in that hour with probability 0.02. The focal multiplexity of each tie, then, is defined as the difference between the multiplexity of the tie and the expected induced multiplexity.

While the expected induced multiplexity increases with contact frequency, the average focal multiplexity does not show the same trend and is rather constant across the contact-frequency range (see figure 4.2 (c)). The focal multiplexity is negative for most of the ties, indicating that ties favour their own social times. Moreover, the focal multiplexity sustains a similar level of variance across the contact frequency range, showing that multiplexity is not merely caused by a high frequency of contacts: both low and high-contact ties can be of varying levels of focal multiplexity.

Another factor which may affect the measured temporal multiplexity of ties is the bursty calling patterns [28, 81]. Many forms of human communication, including mobile phone calls, are bursty, meaning that the communication timeline consists of brief bursts of high activity divided by long periods of inactivity. Moreover, the general level of burstiness can depend on the communication medium as well as on how and in which units the communication activity is being measured. One way to reduce the medium-specific effects is to take a coarse-grained approach for measuring the tie activity and count the number of *active hours* instead of the number of communication events (a method also used in Publication I). We tested the robustness of the measured values of temporal multiplexity by comparing the multiplexity values when the tie activity is measured in

number of calls versus when it is measured in number of active hours. We did this because we were interested in the effects related to the nature of social ties and not the temporal patterns associated with the medium of communication such as different levels of burstiness.

The comparison of the measured multiplexity shows that the impact of burstiness is negligible, and thus, when it comes to quantifying temporal multiplexity, both methods of tie activity measurement can be used (See Figure 4.2 (d) for the comparison). For the sake of simplicity, we used call counts in the rest of the analysis of the Publication IV.

4.3.2 The strength of monoplex ties

Based on Feld’s theory [54], tie multiplexity is associated with alter overlap and tie bridginess (See Figure 2.1 for the definition of alter overlap). In Publication IV, we investigated the interplay between the temporal multiplexity and alter overlap and bridginess of the ties. Our results show that alter overlap increases with temporal multiplexity. To disentangle this effect from the well-studied effect of bridginess of weak ties [63, 116], we grouped ties based on their contact frequency and examined the effect of temporal multiplexity on the alter overlap for each group separately (See Figure 4.3 (a)). We observed that for all groups, the low-entropy ties (the ties which are less multiplex) consistently have lower overlap than high-entropy ties. Our results suggest that even high-frequency ties can be bridging when they have low temporal multiplexity and weak ties are less bridging when their activity, even though small in volume, is dispersed across several social times.

Besides the alter overlap, which gives a measure of the local bridginess of the ties, in Publication IV, we also investigated the contribution of monoplex/multiplex ties to the overall connectivity of the network by studying the growth rate of the Largest Connected Component (LCC) when adding links according to their entropy (which is a measure of multiplexity). Previous studies on networks of mobile phone communication have shown that adding low-contact ties results in a faster emergence of LCC as compared to adding high-contact ties [117] – a phenomenon predicted by Granovetter’s theory on the strength of the weak ties [63]. We use a similar approach to determine the effect of temporal multiplexity on the emergence of LLC. We observed that adding the low-entropy ties first results in faster growth of the LLC size as compared to adding the high-entropy ties first (see Figure 4.3 (b)). This observation is in line with Feld’s theory and indicates that the measure of temporal multiplexity coined in our study can encapsulate the essence of tie multiplexity.

Moreover, we compare the scenarios of adding low/high-entropy ties with the well-studied scenarios of adding low/high-frequency ties first (see Figure 4.3 (b)). We observe that adding low-entropy ties results in a

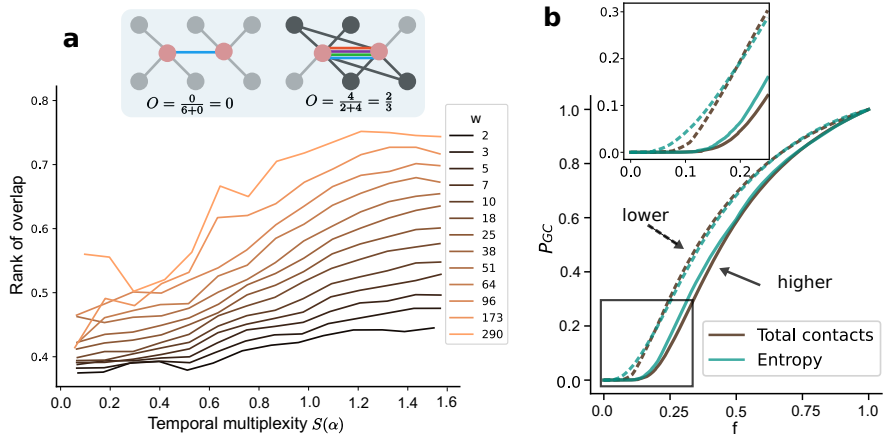


Figure 4.3. Monoplex ties are both locally and globally bridging. Feld’s theory of the focused organisation of social ties predicts that multiplex ties are less bridging. In (a), we checked if this holds true for our measure of temporal multiplicity by investigating the relation between the temporal multiplicity of ties and their overlap (which is a measure of local overlap). (b) We tested the relation between temporal multiplicity and global connectivity by comparing the growth rate of the largest connected component (LCC) while adding the ties to an initially non-wired network (including all the nodes but no ties) according to four different scenarios: 1) adding higher-contacted ties first; 2) adding lower-contacted ties first; 3) adding high entropy ties first; 4) adding low entropy first. We observed that prioritising the low entropy ties (the ties with higher temporal multiplicity) results in the fastest growth rate of the LCC (figure taken from Publication IV).

faster growth rate of the LCC as compared to adding the weak links first, suggesting that monoplex ties may be more crucial for the cohesiveness of the network than low-contact ties.

4.3.3 Egocentric networks and multiplexity

As explained in section 4.3.1, ties favour their own social times and the activities of each tie are temporally focalised. In Publication IV, we also examined whether egos favour social times based on how they allocate their alters in different layers. Moreover, we investigated how the egos’ degree k affects their layer preferences. To figure out if the layers are under-represented or over-represented, for each ego, we determined the *expected alter representation* in the layers under a shuffling scenario similar to the one used in 4.3.1. Figure 4.4 depicts the average ratio of observed alters to expected alters in each layer as a function of ego-net degree k . We observed that the average ratio is smaller than one for all the layers across the degree range, indicating that, in reality, the alters are present in fewer layers as compared with the shuffled scenario. Moreover, we observed that the layer allocation of the alters is degree-dependent. Particularly, larger-degree egos allocate a higher portion of their alters to the weekend-evening (We-E) layer. This result is in line with the observation of Aledavood *et*

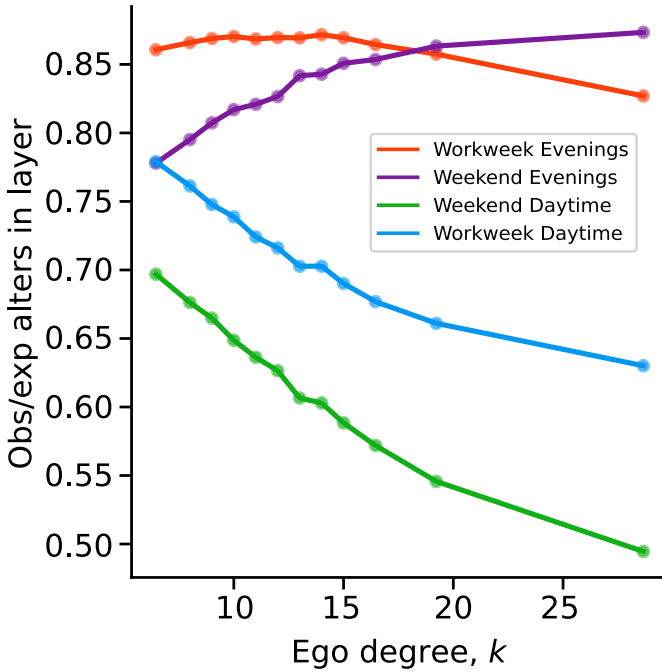


Figure 4.4. Egos with large degree, k , allocate more alters into their Weekend-Evening layer (figure taken from [Publication IV](#)).

al. [3] that *night owls* – people who are more active in the evenings – tend to have larger degrees.

Moreover, in [Publication IV](#), we also investigated the over-representation/under-representation of the triangles around egos in each layer (see Figure 2.1 for a schematic visualisation of triangles in an example network). We calculated the expected number of triangles by averaging over the number of triangles in an ensemble of random configurations of the network (see [Publication IV](#) for the details). After establishing the baseline, we calculated the ratio of the observed number of triangles to the expected number of triangles for each layer. We observed that triangles are more likely to exist in layers that include a smaller proportion of the alters of an ego. This observation can be interpreted according to Feld’s focus theory [54]. Feld hypothesised that large foci (foci involving many people) tend to be less binding, while smaller foci tend to be more involving and encourage interaction among the people. Social times that are used for interacting with only a few alters may correspond to the small and binding foci and thus be used by the alters for communication among themselves and consequently include more triangles. On the other hand, the social times that include a large portion of the alters may correspond to large and less binding foci that do not foster triadic closure, or alternatively, these social times may correspond to several disjoint foci (and not merely

a single focus) and thus do not include many triangles.

Moreover, we studied the over-representation/under-representation of the triangles around egos in the aggregated (collapsed) network with a similar approach. We defined an alter's embeddedness in an ego's personal network as the number of triangles that both the ego and alter belong to. We observed that alters who are contacted at the weekends (i.e. the alters that are present in the weekend evening and the weekend daytime layers) are more embedded into the ego's network. This result is in line with the observation of Krings *et al.* [89] that the weekend communication events are more frequently associated with high-overlap ties and dense clusters. Moreover, this result may indicate that the foci that are active during the weekends tend to be more involved and binding.

4.3.4 Discussion

In Publication IV, we suggested a method for inferring the multiplexity of social ties from their temporal expressions in the absence of explicit information on the social contexts underlying the social ties. It is because of the spatiotemporal regularities of human communication [1] that we can extract meaningful social times from the population behaviour and then categorise each social tie according to these social times. The large sample size of our data makes the detection of meaningful social time signals possible. However, we did not investigate the effects of the demographic composition of the studied population on the inferred latent social signals. Future studies can shed light on the dependence of the detected social times on the demographic composition of the sampled population.

Moreover, our study was based on call logs. It would be interesting to apply the same method on other types of digital traces of human interaction such as other communication logs (e.g. texting, social media activity) as well as proximity data collected by RFID (radio-frequency identification) tags or Bluetooth sensors.

Additionally, our empirical findings are based on a single dataset consisting of call logs of a sample population in an undisclosed European country. Our hypothesis on the temporal expression of social foci needs to be examined against more datasets to investigate the generalisability of our findings across different cultural contexts.

Another direction for future studies is to investigate the nature and implications of the detected weekly social times. For instance, our observations indicate that the ties that use weekend social times are different in some aspects (such as embeddedness) from the ties which use weekday social times. Does the usage of social times affect spreading dynamics? How about the effects of social times usage on the dynamics of social ties? Do social ties change their social-time preferences or their extent of multiplexity as they strengthen or decay? Further studies are needed to answer these

questions.

5. Human Mobility Modelling

Accurate knowledge of human mobility patterns is indispensable for various applications, such as public transportation planning [65, 18], emergency response management [98, 76, 9], and modelling the spread of contagious diseases to enable effective interventions for their containment and control [10, 16, 130, 71, 121, 120, 115].

Mobility patterns of different spatial and temporal resolutions are relevant for each application. For instance, optimising the design of the interior of a busy metro station, a city’s public transport network, or a national flight network each requires different spatial resolution of data on human movement patterns. Similarly, while infrastructure planning often considers long-term and stable trends, natural disasters or pandemics can result in abrupt changes in the patterns of mobility due to both the event itself and the large-scale response interventions. Thus, in such cases, a higher temporal resolution is needed to capture the resulting shifts in the mobility patterns [98, 158, 13, 71]. A recent case of sudden shifts in mobility patterns happened during the COVID-19 pandemic when response interventions such as lockdowns, school closures, remote-work recommendations, and travel restrictions took place [68].

A major approach to modelling the spread of infectious diseases is compartmental metapopulation modelling. The core idea behind this approach is that an infectious disease spreads via two mechanisms: primarily locally and as a result of the intense mixing of people inside densely-populated regions (compartments), and additionally across the compartments when infected individuals travel from one region to another and mix with the people in the destination compartment. Thus, compartmental metapopulation models require information on the aggregated flux of population moving between densely populated regions (e.g. cities or municipalities). Such mobility data can be represented in the form of matrices, which are commonly referred to as *origin–destination travel matrices (OD matrices)* where each entry denotes the number of trips from a geographical area (origin) to another (destination). In Publication V, we investigated the utility of a variety of digital traces and modelling approaches for estimat-

ing the OD matrix in Finland during a time when mobility patterns and travel behaviour were heavily disrupted as a consequence of the COVID-19 pandemic and the classic models of mobility were futile.

5.1 A brief history of OD matrix estimation

An established approach to estimating OD matrices is conducting surveys about people's travelling habits. However, such surveys are prohibitively expensive, not scalable, and prone to recalling bias, as elaborated in chapter 1.

Searching for general laws of human mobility is another long-attended endeavour for estimating OD matrices. Since the 19th century, scientists have recognised that the amount of mobility flow between an origin and a destination is correlated to their populations and inversely correlated to their distance [127, 14]. These two rules are the essence of the *gravity model of mobility* [176]. Another suggested mechanism for explaining the patterns of mobility is *the law of intervening opportunities* proposed by Stouffer in 1940 [154]. Stouffer suggested that the flow of people to a destination is proportional to the number of opportunities in the destination but inversely proportional to the number of intervening opportunities which are located at a closer distance. This idea is the gist of the *radiation model* developed later in the 2010's [145, 128].

Gravity and Radiation models, despite their accuracy flaws [103], are currently the standard methods for inferring mobility from the geographical distribution of population and jobs. These models provide an overall static picture of the mobility patterns but fail to capture the temporal variations such as the weekly and daily rhythms or the sudden and occasional deviations.

Another approach is to infer the mobility patterns from the digital traces that people leave behind. For this purpose, researchers have utilised data such as the registry of smartcards used for getting on public transport (e.g. [75]), the geo-locations associated with the posts shared on social media (e.g. [124]), and mobile phone data (e.g. [14, 91]) collected by both teleoperator companies and smartphone application providers (See section 1.2 for a review of different categories of mobile phone datasets).

5.2 Estimating mobility during disruption

Publication V aims to provide practical solutions to the challenging, yet crucial problem of real-time estimation of volatile inter-regional mobility patterns, with an emphasis on using public data. Real-time insight into mobility patterns is essential for the purpose of crisis management and

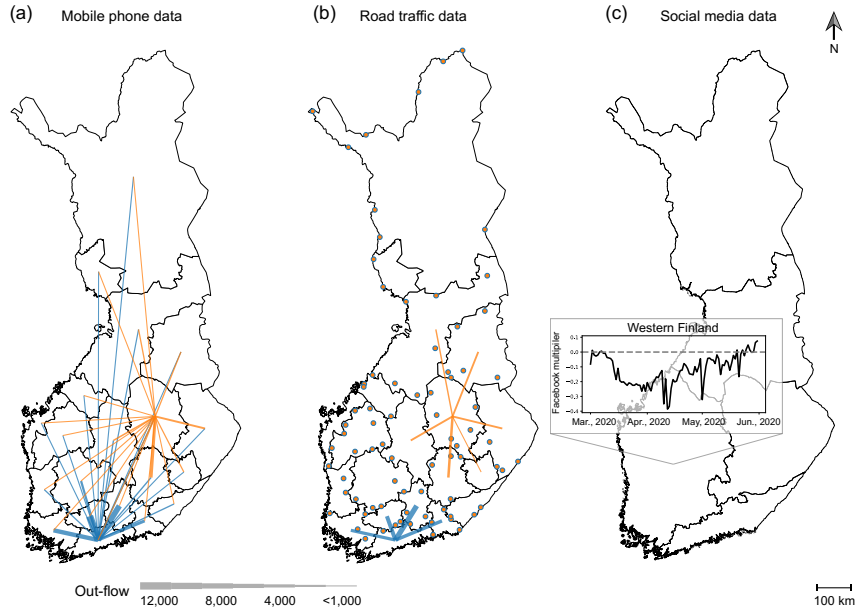


Figure 5.1. Schematic illustration of three types of mobility data used in [Publication V](#). Panel (a) demonstrates example entries from mobile phone data overlaid on a map of 20 Finnish hospital districts. The blue and orange lines each demonstrate the number of trips originating from a pair of districts on a typical day. The thickness of the lines corresponds to the number of trips. While mobile phone data provides a full OD-matrix, the road traffic data (demonstrated in Panel (b)) reports the number of motor vehicles travelling between the adjacent districts. These numbers are calculated from the data recorded by the traffic sensors located near the borders of the hospital districts (marked by dots on the map). Panel (c) demonstrates the Facebook data, which reports how the level of movement of people *inside* each region has changed as compared to a pre-pandemic baseline. Thus, this data, unlike the previous two, does not provide any information on the number of the trips *between* regions. Moreover, the geographical resolution of this data is coarser compared to the other two. The inset graph shows the Facebook mobility change index for an example region from March to June 2020 (figure taken from [Publication V](#)).

the evaluation of the effects of responsive policies. In [Publication V](#), we devised a variety of models, each using different data sources and modelling approaches, and tested their performance in estimating the mobility flows between Finnish hospital regions during the first wave of the COVID-19 pandemic. Even though we took a particular country during a specific duration as a case study, our methods strive to remain general, and the article’s insights can be applied to other cases as well.

Among different digital trace data sources, the location data collected by teleoperators with large market shares plausibly give the most reliable picture of the population movement patterns. However, such data are generally not freely accessible for research purposes and need to be purchased from the teleoperators [[115](#)]. Moreover, even in those cases, the acquired data usually consists of the movement patterns in the past and real-time access to the current mobility flows is not provided. A real-time

insight into the mobility flows is especially crucial for developing timely and adaptive strategies in crisis management.

Meanwhile, publicly accessible road traffic data stands out as a valuable source capable of offering real-time insights into changing mobility trends. This data, collected around the clock through automated sensors, primarily serves to monitor traffic conditions and road utilisation. Unlike mobile phone data, road traffic data does not require anonymisation and is usually freely shared by the traffic authorities. Nonetheless, a shortcoming of road traffic data is that while mobile phone data provides a full OD matrix, road traffic data only depicts a single mode of transportation and is sampled from specific locations along major roadways (See Figure 5.1 for schematic comparison of these two data sources).

Our results highlight the utility of road traffic data in estimating the deviated inter-regional mobility patterns. For instance, we showed that the accuracy of radiation and gravity models are significantly improved when they are augmented by road traffic data. Among different evaluated models, a model based on pre-pandemic mobile phone data and informed with road traffic data during the pandemic performs best in estimating mobility flows. This result points out that even if real-time access to mobile phone data is not possible, we can extract the essential information from outdated data and make accurate real-time estimations by combining this information with real-time road traffic data.

In Publication V, we also investigated the utility of mobility-trends reports published by Google and Facebook for a specific duration during the pandemic (See Figure 5.1 for a schematic demonstration of Facebook data. For a description of the data, see sections 1.2.3 and Publication V). We augmented radiation and gravity models with these publicly available reports. Our results indicated that the static models of gravity and radiation are not enhanced by such augmentation.

Figure 5.2 depicts the weekly estimation error of all the models that we tested during the course of the first wave of the COVID-19 pandemic hitting Finland. See Publication V for a detailed description of each model as well as a summary of the enforced restrictions by the Finnish government resulting in the deviation of the mobility patterns.

5.3 Mobility vectors and social signatures: similar approaches, different applications

In Publication V, we applied the concept of ego-centric networks (a central concept in the other articles included in this thesis and to which Chapter 3 of this thesis is dedicated) to the study of inter-regional mobility patterns. Essentially, for each region r , we made *region-centric* networks consisting of region r in the centre connected via weighted links to the rest of

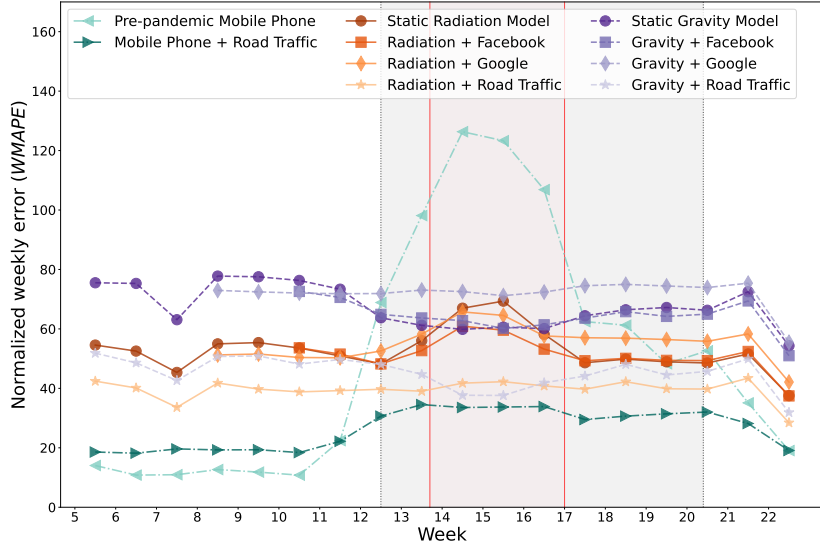


Figure 5.2. Comparison of the weekly country-wide estimation error of the different mobility models devised in [Publication V](#), during weeks 5 to 22 in the year 2020. The pink shades mark the duration in which non-essential traffic to/from the capital region was banned. The grey shades (between weeks 12 and 21) mark the period of imposed governmental restrictions including school closures. Overall, the model combining pre-pandemic mobile phone data with road traffic data has the best performance. The static models of gravity and radiation model that are augmented with road traffic data have the second and third best performances during weeks 14 to 17, which is a duration when the most deviation in mobility behaviour was observed (figure taken from [Publication V](#)).

the regions, where link weights denote the mobility flow between region pairs. From the region-centric networks, we made *mobility vectors* that capture the connectivity profile of each region. Once again, this concept is, in essence, similar to another core concept studied in Publication I and Publication III, namely the concept of social signatures explained in Section 3.4.1, except that the mobility vectors preserve the information about the origin/destination regions, while social signatures do not preserve the information on the identity of alters.¹ The mobility vector of a region encapsulates the way it is embedded in the country’s mobility network. In Publication V we employed mobility vectors in the model that combines mobile phone and road traffic data, as well as in the radiation and gravity models that are augmented by the road traffic data.

5.4 Discussion

To gain insight into the efficacy of different data types, Publication V focused on tractable and relatively simple models for estimating shifting mobility patterns. Our results indicated that the best strategy is to use past mobility patterns (e.g. from previous years’ mobile phone data) as a basis and then enhance them with real-time road traffic data. In summary, in this modelling approach, we did linear regressions to determine the relationship between population flow according to road traffic data and mobile phone data. Moreover, we used mobile phone data to construct temporal mobility vectors for each region. Thus, by having the total amount of vehicle flow from a region according to the road traffic data (which is public data and continuously available), we can estimate the outflow of people based on the coefficients obtained by linear regression. Then, we can use the mobility vectors obtained from the historical mobile phone data to estimate the flow from the origin region to different destinations.

In the lack of historical data on mobility patterns (e.g. pre-pandemic mobile phone data), we suggest using the predictions of either the radiation or gravity model as a basis and then enhancing them in a similar way with real-time road traffic data.

As mentioned earlier, the emphasis of Publication V has been on simple and tractable models. Such models are useful for gaining insight but may be outperformed by more sophisticated models. Future research on the comparison of our models with models relying on machine-learning and deep-learning techniques (such as [173, 99] and [174, 144]) is needed to map and understand the transparency-vs-precision dilemma.

Another topic for further studies is the evolution of mobility vectors

¹Astero *et al* [6] have studied *mobility signatures* which, similar to social signatures, are sorted arrays of fractions and thus do not preserve the information on destination cities (unlike mobility vectors).

over time. Our best-performing model combines pre-pandemic mobility vectors with real-time road traffic data to nowcast the mobility flows. Thus, the model implicitly assumes that the mobility vectors are sufficiently stable. Having assumptions is not a drawback per se, as long as the model works. In fact, as explained in Section 5.1, popular mobility models of gravity and radiation each have their own assumptions on the mechanisms ruling human mobility behaviour. Nevertheless, longitudinal studies are desirable to investigate the evolution of mobility vectors. The mobility patterns may drift over time; thus, using vectors that are too outdated as a basis would reduce the model's performance. Further studies and longitudinal data are needed to understand the rate of such drifts and estimate the expiry date of past data for mobility nowcasting.

6. Conclusion

6.1 Summary

This thesis provides insights into the dynamics and persistence of human behaviour patterns through analysis of digital traces and highlights the importance of computational methods for gaining a quantitative understanding of human behaviour. Our results both contribute to the foundations of computational social science and have potential for practical applications (as demonstrated by the model that estimates mobility flows).

In the publications covered in Chapter 3 (Publication I, Publication II, and Publication III), we identified the individual and universal features of personal social networks. In particular, we used a wide range of communication data to provide evidence on the persistence of social signatures in time (Publication I, Publication II). Moreover, we showed that people have similar-looking social signatures across the call and SMS communication channels (Publication I, Publication II). In Publication II, we highlighted the universal attributes of human ego-centric networks by examining the communication patterns of millions of people via various channels (e.g. mobile phone calls and text messages, email, online discussion forums). Additionally, to explain the generative mechanism behind the observed universal patterns and the individual variations within them, in Publication II, we presented an ego-network model that connects the structure of ego-networks to the individual communication style of the egos. In Publication III, we disentangled the structural properties of social signatures (length and tie strength heterogeneity) and showed that while there exist different distance functions for measuring the similarity of ego-networks, some of them are more suitable for comparing ego-networks of different sizes.

Publication IV (covered in Chapter 3) bridges traditional social science with computational approaches. We used high-resolution call logs to provide evidence for Feld's theory on the focused organisation of social ties

and its implications on structural attributes of ties that are associated with several social foci.

Finally, in Publication V, we demonstrated that an egocentric network approach can help us understand mobility flows better and allow us to make real-time estimations of country-wide mobility flows by combining various digital data sources, even during significant distortions in mobility patterns such as those we experienced during the COVID-19 pandemic.

6.2 Ethical Considerations

This thesis demonstrates the great potential of digital traces for expanding our understanding of human behaviour and for contributing to social good. However, the abundance of digital traces of different aspects of our daily lives has brought not only new opportunities but also new menaces and, consequently, new ethical challenges [136]. The digital traces can be misused, for instance, by totalitarian states for surveillance of the citizens and imposing social control or by companies to manipulate people's behaviour for the sake of companies' interest. However, not all perils are this evident: sometimes even well-intended research can cause harm to people, for example, by violating their privacy [136].

All the presented works in this thesis are conducted by considering the subjects' privacy. The datasets we have used are anonymised and already well-studied in the literature. Moreover, the methods developed in this thesis cannot potentially be used for de-anonymisation of the subjects. For instance, *social signatures* cannot be used for identifying the corresponding subjects (even though the term *signature* might suggest so). The term *signature* is merely used because there exists a certain kind of continuity and persistence in the communication behaviour of people. However, the signatures are not unique and cannot be used to identify an individual from a crowd. Similarly, the mobility model that we have developed in Publication V nowcasts the aggregated population flows between large areas and can not be used for predicting the mobility patterns of individuals.

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